International Journal of Heat and Mass Transfer 183 (2022) 122166

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Thermal properties of phase change materials reinforced with multi-dimensional carbon nanomaterials

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ARTICLE INFO

Article history:
Received 23 June 2021
Revised 24 September 2021
Accepted 19 October 2021
Available online 2 December 2021

Keywords:
Phase change materials
Composites
Thermal properties
Multi-dimensional carbon nano-materials
Thermal stability
Encapsulation
Functionalization
Crystallization
Percolation threshold
Graphene
Thermal conductivity
Latent heat
Enthalpy

ABSTRACT

Energy conservation is an ever-growing field of research, due to the global energy demands continuously escalating and the need of cleaner energy sources with a reduced carbon footprint. To preserve resources, latent heat thermal energy storage systems, utilizing phase change materials (PCMs) are being researched due to their high heat capacities and the lack of practical thermal conductivity. This review focuses on a numerical comparison of retained enthalpy and thermal conductivity enhancement (TCE) of PCMs with additions of multi-dimensionality carbon nanofillers, that are functionalized and encapsulated, whilst trying to account for the physical phenomena that influence the effects. Interfacial chemistry between the filler and PCM will affect filler dispersion, composite TCE, enthalpy and stability. The thermal percolation threshold has been explored for 0D, 1D, 2D and 3D fillers. Assessments for the best dimensionality filler at low weight fraction have also been made. The highest TCE at low weight fractions is seen from 3D carbon nanomaterials, and at high weight fraction from unfunctionalized 2D carbon nanomaterials. The data gathered from many research papers has allowed the confirmation of a thermal percolation threshold in 2D nano carbon filled PCMs. By studying the intermolecular interactions of the filler and PCM material, the composite properties can be maximised, such as achieving a high composite enthalpy via maximizing the degree of PCM crystallization. Therefore, this review will have use for researchers to plan new experiments and head in a forward direction towards innovation. However, there is a need for more research in this field on the topics of 1D and 2D nanocarbon fillers at high wt%, in order to compare their effectiveness and percolation with the other dimensionality fillers. There is, also, a need for more concise and complete reporting of data by researchers in the field due to the complex interconnected influences of the different material properties; doing so will make reviewing and comparison more accurate and faster, allowing more conclusions to be drawn from further review by future researchers.

1. Introduction

Most of today's world is reliant on non-renewable energy sources, creating undesired effects on the natural,economic and political environment [1] which is highly detrimental to the stability of global wellbeing. Energy saving is a large concern in current times, with predictions that a rise in consumption of 48% will occur by the year 2040 [1]. With these rises in demand for energy, more stable solutions should be sought after. Renewable energy sources such as solar, wind, and hydroelectric have been available for many years [2] and they can still be improved upon [3,4]; however they still do not comprise a large part of the global energy production as presented in **Fig. 1**. With predictions that the prices of non-renewable sources of energy will increase due to their scarcity [5], it would be wise to prepare for a future where we must conserve energy and rely on renewable energies, as many countries are currently doing [6]. To do this, new technologies should be devised and/or optimised to facilitate renewable energy usage and energy conservation.

Many areas of energy use require heat removal or containment for energy conservation or for performance improvement. Solar heat storage, building insulation, thermal recycling and many more are areas that could benefit from thermal energy storage (TES) devices [7–9]. Thermal energy storage is based on the principle that thermal energy is absorbed and stored in a medium so that it can later be released and used for other purposes. These storage systems have the potential to reduce the amount of thermal energy waste. Phase change materials (PCMs) have therefore got a lot of attention recently due to their high heat capacity and thermal storage potential as evident from the data presented in **Fig. 2**. Phase

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https://doi.org/10.1016/j.ijheatmasstransfer.2021.122166
0017-9310/

List of abbreviations

BNNT	Boron nitride nanotube
BNNS	Boron nitride nanosheet
CNF	Carbon nanofiller
CNT	Carbon nanotube
CTAB	Hexadecyltrimethylammonium bromide
CVD	Chemical vapour deposition
EmPEG	Epoxidized methoxy polyethylene glycol
f-GNPs	Functionalized graphene nanoplatelets
FLG	Few layer graphene
f-MWCNT	Functionalized multi-walled carbon nanotube
GNP	Graphene nanoplatelet
GO	Graphene oxide
HDA	Hexadecyl acrylate
Impreg	Impregnation
MCMS	Mesoporous carbon microspheres
MWCNT	Multi-walled carbon nanotube
NDPCM	Nano-dispersed phase change material
OA	Oleylamine
OI	Octadecyl isocyanate
PCM	Phase change material
PUB	Polyurethane binder
Ref	Reference
rGO	Reduced graphene oxide
SDBS	Sodium dodecyl benzene sulfonate
SDS	Sodium dodecyl sulfonate
SEM	Scanning electron microscopy
SWCNT	Single walled carbon nanotube
TCE	Thermal conductivity enhancement
TES	Thermal energy storage
TGA	Thermogravimetric analysis
xGNP	Exfoliated graphene nanoplatelet
XRD	X-ray diffraction

Symbol definitions

A	Particle shape coefficient
c	Speed of sound (m s^{-1})
k_{eff}	Effective thermal conductivity (W $m^{-1}K^{-1}$)
K_{eff}	Effective thermal conductivity of the material (W $m^{-1}K^{-1}$)
t_{AB}	Transmission coefficient of normal phonon energy between material interface A to B
v	Volume fraction
w	Mass fraction
λ	Conductivity ratio
ρ	Density (kg m^{-3})

Subscripts

A	Material A
B	Material B
C	Composite
f	Particle/filler
m	Matrix
M	Maximum filler fraction

Units

µm	Micrometres
nm	Nanometres

change materials are materials that have a high specific heat capacity, and latent heat capacity which is the measure of heat capacity associated with a phase change. An interesting area of application is the enhancement of the cycle life of Li-ion batteries, which can be improved by managing the temperatures of the cells closely, using composite phase change materials (CPCMs) [10,11].

The majority of current PCMs do no exhibit all of the key properties (see **Section 2. Desirable thermal properties of phase change materials** for more details) needed for useful and effective thermal energy storage systems, therefore enhancements need to be made via compositing the PCM with other materials, such as making composite phase change materials (CPCMs) [12,13]. Some methods that are used to improve the conductive and convective heat transport in PCMs are displayed in a hierarchical graphic in ***Fig. 3***.

In regard to reviews and reports of PCMs here has not been a review detailing, comparing, and explaining the effective properties of different dimension carbon nanofillers, each with or without surface functionalization or encapsulation. Few review papers even directly compare the effectiveness of each filler (see **Table 1**) whilst trying to account for physical properties (such as dimension), surface modifications in addition to the weight percent (wt%) of filler and its thermal conductivity. Providing ranked thermal conductivity enhancements (TCE) and graphs showing the effects of different methods for each kind of carbon nanofiller will make it easy for other researchers who are trying to understand how to optimise the thermal properties of PCM; this is the main aim of this review report.

For researchers seeking to use CPCMs for thermal regulation applications, it is most likely that they will have to conduct their own costly, and time-consuming research so that the material suits their application well. Due to the many variations in properties and effects and the different needs for a certain application it would be useful to understand and predict the property outcomes of the final CPCMs; doing so would open up possibilities to others that were not considered before. This research will focus on clearly distinguishing the effects that different types of carbon nano filling particles will have on the thermal properties of PCMs and will try to explain the influence that the physical phenomena have on them.

2. Desirable thermal properties of phase change materials

For the phase change materials to be suitable for a specific application there are many properties that must be considered before assuming a choice of material. A necessity is that the PCM material must be cost effective, safe and easily obtainable [5]. The following sections will discuss commonly sought-after properties.

2.1. Thermal conductivity

For a PCM to be useful in a thermal energy storage system it has to have a fast thermal charging and discharging cycle, which is determined by the composite thermal conductivity; this is so that the heat is not dissipated elsewhere and is absorbed by the PCM. Often the composites have a thermal conductivity which varies dependent on the state it is in [21].Conductive heat transport is dependent on the thermal conductivity of the constituent materials, and the geometries they form either in solid, transitional, or liquid states. Convective heat transport of a PCM is a contributing factor to the thermal conductivity of the composite whilst it is in the liquid state [22]; therefore, if the composite is high in viscosity there will a reduction in the overall thermal conductivity in the liquid state [23–25]. Hence, a low viscosity CPCM is desirable to promote the heat transport and also to improve cycling rates.

2.2. Heat capacity

Specific heat capacity is defined as the amount of heat that will increase a unit mass of materials temperature by one unit; it is not associated with a phase change [26]. A high specific heat capacity

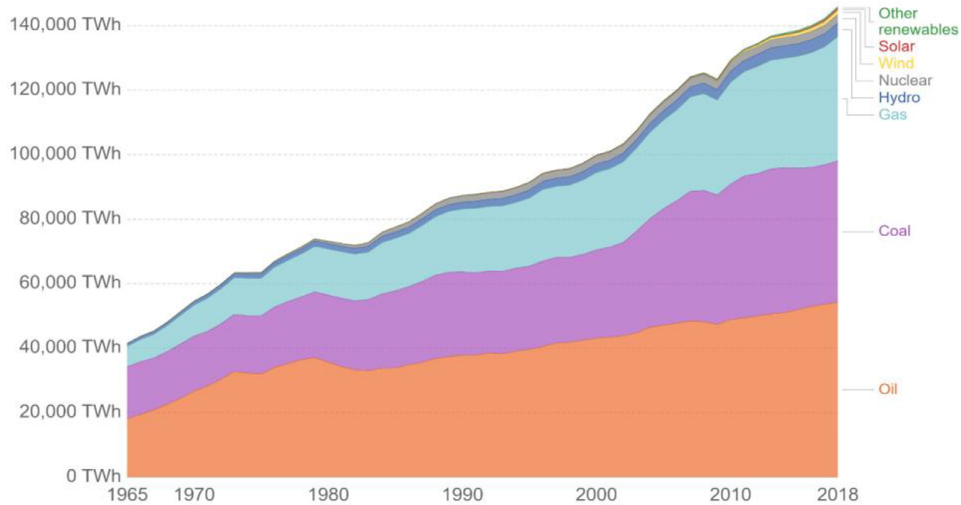

Fig. 1. Graph of global energy consumption by source, in terawatt-hours (TWh) vs year [2,3].

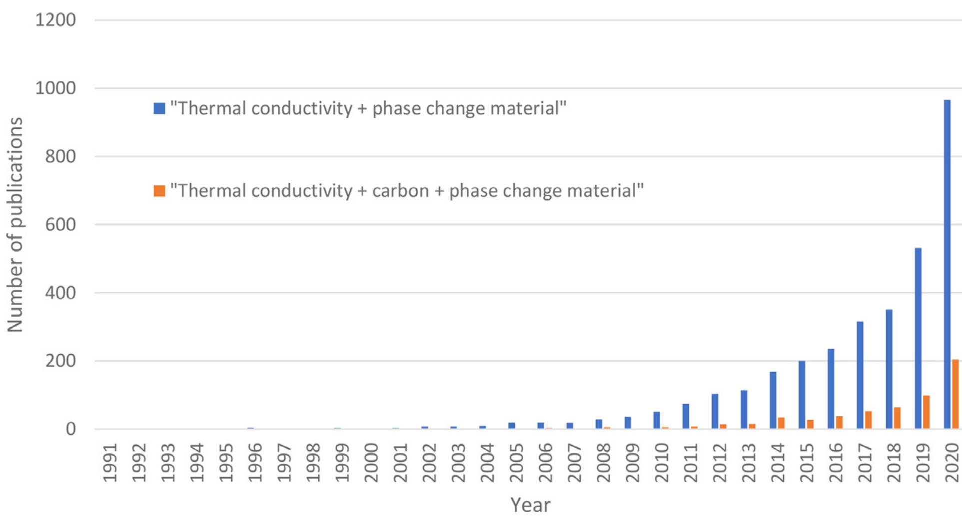

Fig. 2. Search terms and their results from 'web of science'.

Table 1
Previously published review papers on relevant topics.

Author [Reference]	Review contents
Sarbu et al. 2018 [9]	Paper reviews thermal energy storage technologies/systems for utilizing solar heat, reducing building energy usage, and properties of different types of PCM and their encapsulation.
Yang et al. 2019 [14]	This review focuses on the applications and properties of carbon filled PCMs for thermal storage.
Jebasingh et al. 2019 [15]	Review of the physical phenomena that alter the thermal properties of nano dispersed PCMs.
Gariboldi et al. 2019 [16]	This work provides an in depth look at the methods of characterization available for composite PCMs
Lin et al. 2018 [17]	A review on the methods of enhancing the thermal conductivity of PCM materials with different methods such as using additives or encapsulation.
Wu et al. 2020 [18]	Reviews of different dimension filler materials and the effect phonons have in thermal transport in composite PCMs.
Pietrak et al. 2015 [19]	Reviewing the models proposed for enhancing the effective thermal conductivity of composite materials
Nazir et al. [20]	Reviewing the classification and selection criteria of PCMs.

is desirable for PCMs so that the heat storage before and after the phase change is large.

Latent heat capacity is the heat required to change the phase/state of a material at a constant temperature. Latent energy storage systems store thermal energy by undergoing a phase transition at a constant or near constant temperature [27]. A high latent heat capacity will allow more energy to be stored per unit and will reduce the amount/of PCM required for a given application.

2.3. Operational stability

The operating temperature of a PCM is a property that is tuned by the choice of the materials and should be chosen carefully to suit the application of the TES system. The type of PCM mostly influences the phase transition temperature. For those reasons, a composite must have thermal stability and should not deviate significantly from its operating temperature.

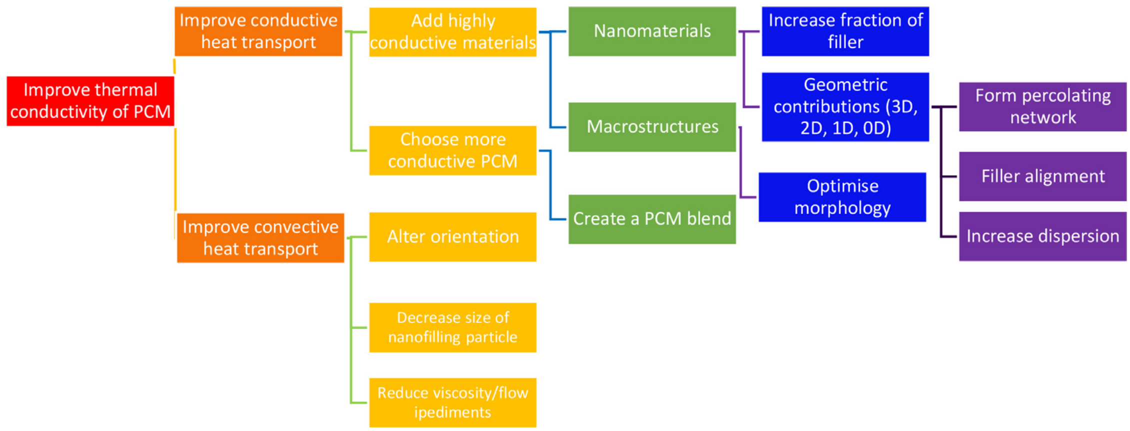

Fig. 3. Hierarchy of thermal conductivity improvement.

The degradation of a PCM material will also occur at elevated temperatures, causing the composite to lose mass and thermal properties. Raising this temperature is desirable, as a safety factor, to extend the range of temperatures at which the system can be employed.

After numerous heating and cooling cycles CPCMs tend to have a reduction in their thermal properties such as heat capacity and thermal conductivity [28]. For most of the CPCMs, reductions in thermal conductivity after cycling can be attributed to the agglomeration of additives such as nano-fillers [24]. Another issue can occur which is when the PCM materials may begin to flow and exit the composite structure, described as leakage, which results in a reduction in the thermal properties of the composite and is thus undesirable [29].

For practical use of CPCMs the durability must be considered. After many thermal cycles phase segregation of the PCM and filling phase may occur which can result in a reduction of latent heat capacity, melting temperature, and thermal conductivity [30]. Typical studies conduct experiments of around 1000 to 2500 cycles [30], followed by a measurement of the composite's properties to measure the change in the thermal properties, which can result in a reduction in the latent heat of solidification value by 1.73% at 2500 cycles in a graphene oxide/paraffin CPCM [31]. When considering practical use, the container that holds the CPCM and the environment in which it is situated must be carefully thought about; these external factors may influence the durability of the composites. Also, a recent review paper which explores the cycling behaviour states that most CPCMs maintain their physical and thermal properties after thousands of heating and cooling cycles [32].

2.4. Chemical and physical properties

For safety reasons, PCMs should be non-flammable, non-toxic, non-corrosive and inert in situ [33]. Its long-term stability must be such that it is economically and environmentally sustainable/viable [34,35]. A low vapour pressure (at the operational temperature) and low thermal expansion coefficient will reduce the chance of container rupture [5]. These properties need to be considered before a specific PCM is chosen for an application.

3. Physical phenomena that effect the thermal properties

The cause of the CPCMs thermal properties can be pinpointed to the microstructural properties and the interactions of the constituents at the nanoscale level; in order to achieve the best results, the causes of them should be understood. The thermal conductivity of a CPCM does not only depend on the inherent thermal conductivity of the base materials. Geometry, fraction, interfacial properties, chemistry and the vibrational modes/phonon propagation through the materials all contribute to produce an overall thermal conductivity value. It is important to understand these effects in order to harness the largest thermal conductivity enhancement possible, therefore several models that seek to represent these behaviours are listed below.

The mass fraction of high thermal conductivity filler particles can have a significant influence on the thermal conductivity of composite PCMs. To convert from fraction to weight fraction for conversion of the filler loadings, Eq. (1) can be used:

$$\boldsymbol{v} = \frac{w\boldsymbol{\rho_m}}{\left(\boldsymbol{w\rho_m} + (\mathbf{1} - \boldsymbol{w})\boldsymbol{\rho_f}\right)} \tag{1}$$

Rearranges to:

$$w = \frac{\rho_f}{\rho_f + \frac{\rho_m}{v} - \rho_m}$$

Where: w is the filler mass fraction ratio, ρ_m is the matrix density, ρ_f is the filler density, and v is the fraction of filler.

There are many mathematical models that have the ability to predict the thermal behaviour of composites containing nanoparticles, however some do not consider critical characteristics such as the shape of the particles, making them less accurate in their predictions. The simplest is the rule of mixtures for series and parallel components [36] however this does not predict percolation behaviour because it does not account for particle shape.

Effective medium theory (EMT) given in Eq. (2) can be used to model the properties of composites.

$$\boldsymbol{v_m}\frac{\boldsymbol{k_m} - \boldsymbol{k_{eff}}}{\boldsymbol{k_m} + 2\boldsymbol{k_{eff}}} + \boldsymbol{v_f}\frac{\boldsymbol{k_f} - \boldsymbol{k_{eff}}}{\boldsymbol{k_f} + 2\boldsymbol{k_{eff}}} = 0 \tag{2}$$

Where v_m and v_f are the fraction of matrix and filler, respectively, k_m and k_f are the thermal conductivity of the matrix and filler and k_{eff} is the effective thermal conductivity. The EMT is modelled after a random dispersion of secondary phase particles [37]. The composite is treated as a macroscopically inhomogeneous medium whose thermal conductivity varies in space [38].

The Maxwell-Eucken model provided in Eq. (3) can be useful in calculating the thermal conductivity of a composite material

$$\boldsymbol{k_c} = \frac{2\boldsymbol{k_m} + \boldsymbol{k_f} + 2\boldsymbol{v_f}\left(\boldsymbol{k_f} - \boldsymbol{k_m}\right)}{\boldsymbol{k_m} + \boldsymbol{k_f} - \boldsymbol{v_f}\left(\boldsymbol{k_f} - \boldsymbol{k_m}\right)}\boldsymbol{k_m} \tag{3}$$

where k_c, k_m and k_f are the composite, matrix, and filler thermal conductivities respectively, and v_f is the filler fraction. The Maxwell-Eucken model [39] is based on a heterogeneous composite composed of a continuous matrix (the PCM) mixed with a dispersion of small spheres (nanoparticles). The spheres are spaced

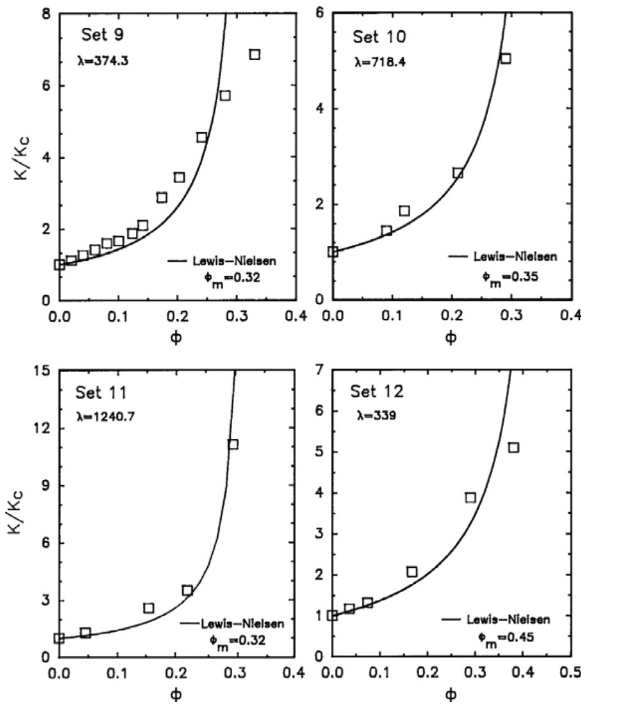

Fig. 4. Predictions of the Lewis-Nielsen model for relative thermal conductivity vs. filler fraction [40]. Where K is the composite thermal conductivity and K_c is the matrix thermal conductivity. Sets 10–12 are made using Graphite particles and set 9 uses Aluminium powder.

so that they do not form percolating networks and are not influenced by each other's temperatures. This model would be suitable for nanoparticles that are well dispersed and at a low fraction.

Lewis-Nielsen model which is given in Eq. (4) is also quite popular in predicting the behaviour of fillers and also simple to use.

$$\boldsymbol{k_{eff}} = \frac{1 + \boldsymbol{AB}\boldsymbol{v_f}}{1 \; - \; \boldsymbol{B}\boldsymbol{\psi}\boldsymbol{v_f}} \tag{4}$$

where

$$B = \frac{k_p/k_m - 1}{k_p/k_m + A}$$

and

$$\psi = 1 + \upsilon_f\left(\frac{1 - \upsilon_M}{\upsilon_M^2}\right)$$

Where v_M is the max filler fraction, v_f is the filler fraction, A is the shape coefficient for the filling particles and k_{eff}, k_p and k_m are the effective thermal conductivity of the composite, filler, and matrix, respectively. The Lewis-Nielsen model [19] considers the maximum packing fraction, although it assumes the particles to be spheres, it does approximate thermal conductivity well when compared to the experimental data [40].

Fig. 4 illustrates the predictions made by the Lewis-Nielsen model along with some experimental data for different types of filler particles by Pal [40]. Where λ is the ratio of thermal conductivities, filler to matrix, Φ_m is the maximum packing fraction of filling particles; as its value is reduced, the size of the particles will tend to increase (size distribution of length, thickness etc.). Therefore, it can be seen that as the particle size increases the abrupt increase in thermal conductivity happens at lower filler fractions. The gradient increase (or thermal percolation threshold) occurs at around 0.15–0.25 fraction for the graphite fillers. Each type of graphite filler (sets 10–12) will have different dimensions, interactions with the matrix and agglomeration properties, giving rise to the different percolation values [40].

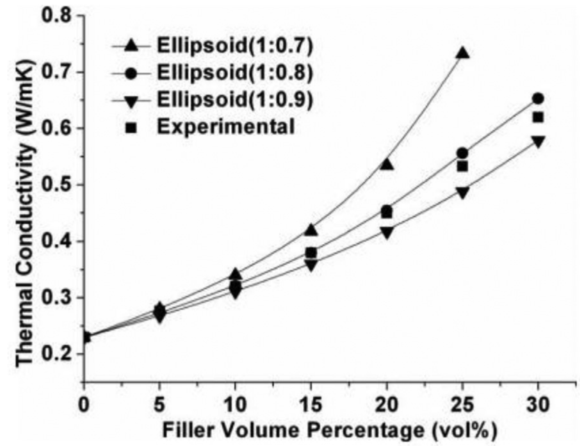

Fig. 5. Simulated thermal conductivity of ellipsoidal particles of differing aspect ratios at different fractions [41].

Fu et al. [41] numerically investigated the effect of fraction of an ellipsoidal particle (with a high thermal conductivity) using a matrix with a low thermal conductivity. Their experimentally measured thermal conductivity results were in good agreement with the predictions made by the Lewis-Nielsen model, showing percolation behaviour at roughly 0.15–0.2 particle fraction. Their work also predicted the effect of aspect ratio on the composite thermal conductivity; that being the thermal conductivity will increase as the aspect ratio increases for the same fraction of filler, as seen in **Fig. 5**.

Conduction is not the only mechanism of thermal transport within a PCM composite. When a PCM is at liquid state, the temperature differences in the melt will cause natural convective forces due to the buoyancy difference. Xiaoquin et al. [42] studied the effects of convection on the heat transfer of a PCM material, they found that increasing the inclination angle of a panel from 30° to 90° containing melted PCM enhanced convective forces and thus accelerated melting. A few other works [43,44] have also claimed that the orientation of the container will affect the heat transfer rates of CPCMs.

Convective heat transport will influence the thermal conductivity value of CPCMs primarily when they are in the liquid state. Work by Hwang et al. [45] on the study of the natural convection of Al_2O_3 nanoparticles in a fluid shows that the convective heat transport of the base fluid can be improved by additions of nanoparticles. This heat transport enhancement (with nanoparticles) is dependent on the fraction (or weight fraction) and the aspect ratio of the particles, where higher fractions and lower aspect ratios will lead to higher rates of convective heat transport. This has also been confirmed by others using copper nanoparticles [46] who also states that the shape of the enclosure will influence the degree of convection. It is also important to note that thermal gradients in a fluid will lead to convective heat transport [26].

Brownian motion describes the random motion of particles dispersed in a liquid, which are subjected to random motions imparted on them by collisions with the high velocity liquid molecules [47]. These random motions and collisions increase the thermal transport and convection giving rise to macroscopic thermal changes in nanofluids. You et al. [48] argues that as the nanoparticle size decreases, the liquid thermal conductivity increases due to Brownian motion, affecting the nanoscale convective transport of the particles as shown in **Fig. 6** displays the effect of reducing the nanoparticle diameter, and also shows the previous

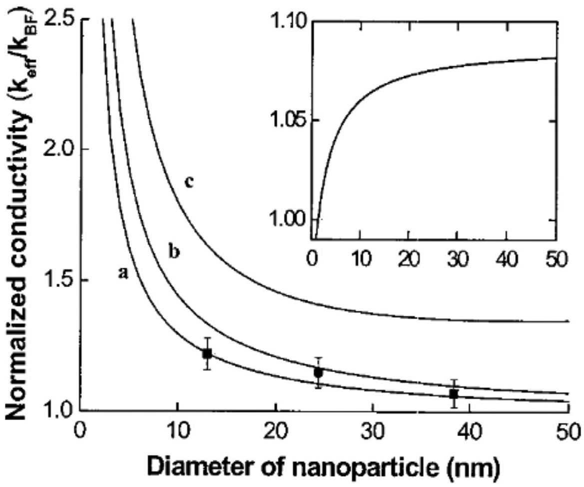

Fig. 6. Liquid thermal conductivity vs nanoparticle diameter: (a) Al_2O_3 in water, (b) CuO in ethylene glycol, (c) Cu in ethylene glycol. With inset being the predictions by the Maxwell model.

predictions by the Maxwell model (graph inset) which are contradictory. Moreover, a number of recent works [45,46,49] support the convective TCE by smaller particle dimensions theory of You et al. [48]; however others indicate that Brownian motion has a lesser effect on the thermal conduction at lower temperatures [50].

It can be assumed that high filler fractions and fixed networks/foams will impair the convection of a PCM melt; there will be a competing effect on the thermal conductivity caused by this and the increase caused by the increase in the path length of the highly thermally conductive filler. However, increased convection and cycling can lead to sedimentation and aggregation of filler particles ultimately reducing the composite thermal conductivity over time.

The agglomeration of nanoparticles is controlled by the Van der Waals attractive forces between them. The average size and stability of the agglomerates will be influenced by the original size of the nanoparticles before they agglomerate; where smaller particles create smaller and more stable agglomerations and vice versa [51]. Leakage of PCMs from the composite is an undesired quality and can be prevented by increased interaction with the filler, and it can be done via increasing the capillary force action [52]. Rule of mixtures can be used to predict the composite latent heat. In most circumstances the latent heat of the filling particle will be zero because it does not undergo a phase change Eq. (5).

$$\boldsymbol{\Delta H = \Phi_p \Delta H_p + \Phi_m \Delta H_m} \tag{5}$$

where ΔH is the composite latent heat, ΔH_p is the filler latent heat, ΔH_m is the matrix latent heat, Φ_p is the filler fraction and Φ_m is the matrix fraction

3.1. Phonon transport

The mechanism behind thermal transport in materials is primarily due to acoustic vibrations that are a product of crystal lattice vibrations, also known as phonons. The mean free path (MFP) of a phonon describes the average distance a phonon will propagate for before experiencing a scattering event; higher MFPs result in higher thermal conductivities [53]. Nanostructures behave significantly differently to bulk materials in their phonon propagation and thermal qualities.

There is a finite thermal resistance at boundaries between different materials caused by a break in the regular crystalline lattice that the phonons are propagating through. Interfaces that exists between two different materials have a resistance which can be explained by acoustic mismatch; if the density and speed at which sound travels in the two materials is very different then there is an acoustic mismatch and thus there will be an interfacial thermal resistance which can be modelled by Eq. (6) [54].

$$\boldsymbol{t_{AB}} = \frac{4\boldsymbol{Z_A Z_B}}{(\boldsymbol{Z_A Z_B})^2} \tag{6}$$

where t_{AB} is the transmission coefficient of the phonon energy in material A incident normal to the interface with material B, and $Z = \rho$c, where ρ is density and c is speed of sound (Cahill et al. 2002 [54]).

There are three types of phonon scattering mechanisms that have the effect of reducing the thermal conductivity of a material. Phonon-phonon, phonon-boundary and phonon-defect scattering are all events which scatter the propagating phonons, creating a thermal resistance within the material [18]. If the introduction of scattering sources to the material results in coherent phonon scattering then the thermal conductivity will not be affected and incoherent scattering of phonons will result in a thermal conductivity decrease [55].

For thermal transport in few layered graphene (FLG), the thermal conductivity will decrease as the number of layers, n, increases; this is because the phonon dispersion changes and introduces more phonon scattering [56,57]. In FLG with $n < 4$, the phonons do not have cross plane velocity so are less likely to participate in phonon boundary scattering events; in graphene with more layers there is increased boundary scattering due to cross plane phonon velocities [56].

Imperfections existing in the material, such as grain boundaries and point defects will scatter phonons, reducing the thermal conductivity. A nanoparticle which is under strain will affect the propagation of its phonons; the dispersions and the rate of scattering will be affected [58]. The thermal conductivity is therefore dependent on the strain condition of a nanomaterial. It is seen that the thermal conductivity of Single walled carbon nanotubes (SWCNTs) and graphene will decrease as the compressive strain on them increases, but for SWCNTs a small amount of compressive strain will enhance the thermal conductivity; at higher compressive strains the particles buckle, increasing phonon-phonon scattering [59]. Graphene experiences maximal thermal conductivity at 0% strain.

3.1.1. Phonon-phonon scattering

Umklapp scattering occurs in crystalline materials and explains the process by which two phonons can interact to form a single phonon. There are two processes by which Umklapp scattering can occur, the U and N processes. In the N process the total phonon momentum is conserved and in the U process the phonon momentum is changed and may allow the wave vector to change direction [60]. The thermal resistivity that graphene-like materials exhibit is inhibited by three-phonon scatterings, which influences the lattice thermal conductivity near room temperatures [60]. In graphene the out of plane phonon mode, (ZA), or flexural mode has been shown to hardly contribute to the lattice thermal conductivity (k_L), reducing the value itself. The lattice thermal conductivity will decrease as the number of layers of graphene increase and will approach the k_L of graphite after five layers [61]. Elastic phonon-phonon scattering primarily contributes to a lowering of the thermal conductivity in graphene [62].

3.1.2. Phonon-boundary scattering

Phonons that travel between interfaces or boundaries in materials, or between materials will be scattered; boundary scattering between materials is due to the acoustic mismatch of the two materials. Two types of phonons scattering can occur at the boundary; specular reflection which is mirror-like and diffuse which scatters

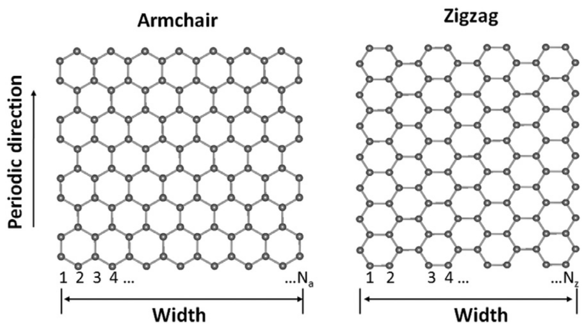

Fig. 7. Different morphologies of graphene; armchair and zigzag [67].

the phonon in different angles. These types of scattering hinder the phonon propagation and thus heat transfer [18,63].

Phonon scattering occurs much more frequently in materials which have larger interface densities [64]. Phonons with lower frequencies tend to undergo stronger scattering events [63]. As the material dimensions become closer and smaller than the mean free path of the phonons, the phonon transport varies and will deviate from Fourier's law [65]. Improving the acoustic mismatch at the interface between the PCM and filler is thus desirable. Attempts to functionalize filling particles with molecules to increase their dispersion should consider the acoustic mismatch model (provided in (6) so that the thermal conductivity is not reduced.

Fig. 7 shows the morphology of zigzag and armchair graphene; the armchair type graphene nanoribbon experiences a lower thermal conductivity when compared to a zigzag nanoribbon of similar dimensions, this suggests armchair graphene nanoribbons have edge effects [66].

3.1.3. Phonon-defect scattering

Defects in the lattice structure of carbon allotropes such as impurities, dislocations and grain boundaries can decrease the thermal conductivity of a material via phonon scattering [68]. However, phonons may not actually be deflected and instead are trapped at the defects where the phonon oscillates for many periods and ultimately decays [69].

3.2. Geometry and percolation

Dimensionality of the filling nanomaterial is often the most important factor to consider when enhancing the thermal conductivity of a PCM with nanomaterials, due to the fact that the long-range thermal conductivity and phonon propagation with the filler is highly dependent on the geometry and dispersion [66]. Aspect ratio is defined as the ratio of length to diameter (or thickness), typically in a fibre, but can be useful for 2D materials such as graphene (length to thickness). The aspect ratio of a filler is a useful factor to have as it will give insight into the degree of percolation possible, or length of phonon path, for a given weight percent of filler used. For example, a 1 wt% MWCNT filler of aspect ratio 1000 will exhibit higher TCEs than a 1 wt% MWCNT of aspect ratio 10 (if all other factors are identical; dispersion, interfaces, and matrix, and dispersion is good) [30]. Yu et al. [72] report an increased TCE with larger aspect ratios. Higher degrees of filler alignment will aid in achieving percolation at lower loadings as demonstrated by Shuying et al. [73]. Methods proposed by We et al. [73] involves applying an electric field to align the filler particles. There might be value in creating a device which will apply or remove an electric field upon energy saturation of a PCM, lowering thermal conductivity so that the thermal energy can be stored until later where the field is re-applied, thus dumping the thermal energy at a moment it is needed.

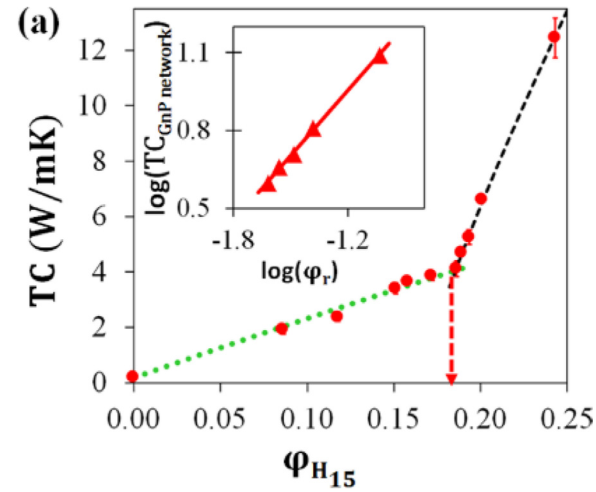

Fig. 8. Thermal conductivity vs. fraction of GNPs with the red arrow showing the thermal percolation threshold [64].

Tuba et al. [70] found that by increasing the aspect ratio of multiwalled carbon nanotubes (MWCNTs) in a high density polyethylene (HDPE) matrix would increase the thermal conductivity by 97% at a maximum of 10 wt% loading of high aspect ratio fillers, vs 63% for lower aspect ratio CNTs. Their explanation for the increase is that higher aspect ratio MWCNT favour the creation of a percolated connected network that supports longer phonon mean free paths. Also, that with lower aspect ratios, the surface interaction with the matrix increases per unit distance. Since there are few studies on the effects of filler aspect ratio on thermal conductivity in a PCM matrix, it would be of use to explore this idea. Although their paper focuses on the effect in a HDPE matrix, the same type of results should be expected in a PCM matrix as long as the morphology of the MWCNT, dispersion and the chemical interactions are the same; HDPE does have a very similar structure/chemistry to paraffin.

Hasan et al. [74] found that increasing the alignment and crystallinity of the PCM phase will increase the thermal conductivity of the composite by 66% in the direction of the aligned molecules of a solid composite with CNT and 87% with graphene; compared to 48% and 52% increase in the liquid phase, relatively [74]. The increase in thermal conductivity is caused by the increase in ordering by crystallization, and the fillers provide a site for alignment of PCM molecules [74]. A needed consideration is that the alignment of the CNTs will lead to an anisotropy in the thermal conductivity; this may limit the application of this finding due to more research being needed to design physical containers that account for this anisotropy.

Percolation roughly occurs at a fraction of 0.17 when considering GNPs [64] (see **Fig. 8**), however this will be different with different types of fillers, mostly due to their aspect ratio/length. Below the fraction of 0.17, there will not be a continuous network of GNPs that permeates the matrix and low conductivity will fill the spaces between the highly conductive filler particles, increasing phonon scattering. At fractions higher than this transition value, there will be a thermally conductive path for phonons to propagate through, with reduced scattering, and fewer high thermal resistance matrix-filler interfaces along the conductive path. These physical effects lead to increased thermal conductivity per wt% of filler used when above the threshold and is thus desirable for improved properties and reduced costs.

Fariborz et al. [75] experimentally and computationally investigated the thermal conductivity percolation in graphene-polymer composites, finding that it is a less defined point and more gradual when compared to electrical percolation thresholds. They found that for graphene fillers a thermal percolation threshold is reached at 30 vol.%. Their explanation for the discrepancy between the electrical and thermal percolation loading fraction is that there is a ‘cross-plane’ thermal conductivity between 2D filler particles, and also that at low loading the thermal transport is dominated by in-plane thermal conductivity of the filler and the intrinsic thermal conductivity of the polymer matrix.

Nandy et al. [76] used weight fractions of graphene in the PCM – ‘RT HC 22′, ranging from 0 to 0.3wt% in increments of 0.05. A linear increase in thermal conductivity was observed, which is congruent with the results reported by Fariborz et al. [75] who explains that below the thermal percolation the behaviour is linear. Others [71] have also argued a linear response in TCE for increased fraction of graphene, although only additions up to 10vol% fractions were tested, not reaching the threshold value reported by Shtein et al. [64]. Shahil and Balandin [71] did ensure a stabilised dispersion of graphene with surfactant methods, therefore percolation was likely possible and not susceptible to agglomeration. Yu et al. [72] reported a linear TCE vs. loading relationship, however their efforts made to sustain particle dispersion was likely ineffective as only sonication was used in an attempt to disperse the nanoparticles.

The experiments performed by Fariborz. et al. [75] found that thermal percolation thresholds in graphene-epoxy/boron nitride-epoxy composites occurred at roughly the same loading fractions for the graphene and boron nitride fillers despite the fillers having a difference in magnitude of 5 in their thermal conductivity values. This can be explained by the geometrical properties of the fillers, as they are similar. The thermal percolation threshold must therefore be highly dependent on the thickness, length, width, and flexibility of the filling material. A less flexible filler material will result in worse coupling to the matrix material and thus poorer thermal transfer across the interface [75].

3.3. Interfacial chemistry between nanomaterial and PCM

Since the surface area of nanomaterials is very high it is favourable for optimal thermal conductivity to minimise the interfacial thermal resistance with the matrix material [70]. Fillers with larger surface areas will interact more with the PCM matrix, providing more opportunities for phonon-boundary scattering and a lowering of the phonon transmissions energy due to the acoustic mismatch of filler and PCM.

However, the interfacial thermal transport of graphene/PCM matrix can be increased by covalent functionalization of the graphene [77]. Longer chain covalent groups will increase van der Waals forces, increasing the interface coupling, and thus improving the heat transfer. Pristine graphene with stearic acid gave the largest reduction in interfacial resistance, due to enhanced coupling caused by hydrogen bonds [77]. Rigid carbon fillers may reduce the interface contact with a PCM by excluding which the PCM may occupy [75].

Doped graphene CPCMs may be a useful way to enhance the interfacial properties of graphene-matrix as shown by Li et al. [78,79], who was able to increase thermal conductivity and aid the PCM crystallization, thus increasing the latent heat. At larger wt% of filler the particles tend to agglomerate, forming aggregates of nanoparticles unless interaction with the matrix is sufficient enough to promote dispersion [80]. Agglomeration of particles is undesired because it reduces the ability of the filler to transfer heat through the PCM and leads to sedimentation of the particles. Functionalization of the filling materials with a molecule whose polarity is the same as the matrix will enhance and encourage dispersion [81].

Capillary forces [82] act to draw liquids and fill volumes, when the pore size is > 50 nm the transport of liquid is faster and for higher porosity volumes is less tortuous [52]. Pore sizes > 50 nm will therefore aid in promoting better infiltration of a PCM into a 3D porous network, creating increased interfaces between the filler and PCM and higher enthalpies per unit of composite. Increased capillary forces will also counteract the leakage of liquid PCM at elevated temperatures, leading to more form stable composites. However, if the pore sizes are too large the capillary forces on the PCM will be smaller and the composite will not act to retain the material; Nomura et al. [83] states that a pore diameter below 10 μm will allow expanded pearlite to retain erythritol well.

The polarity of both the filling material and the matrix/PCM influence the behaviours the of the composite; if both have the same polar nature then the interactions will be enhanced and the composite will exhibit better stability in terms of the degree of dispersion of the filling particles [84].

4. Phase change materials

4.1. Types of PCMs

Energy storage systems can be classified by the physical changes that occur when a heat exchange happens. A sensible heat storage materials energy storage performance depends upon the heat capacity and temperature change of the material during charging and discharging [85] and will only occupy a single phase during its operation. They theoretically have an unlimited number of thermal cycles over their lifetime. Latent heat storage materials have the ability to absorb or release energy during a phase change at isothermal conditions and can reversibly do so; using a material that undergoes a phase change in its operational temperature range enhances the energy storage density when compared to sensible heat storage materials [14]. Table 2 displays the melting temperatures and latent heats of different types of latent heat storage materials.

4.1.1. Inorganic PCMs

Inorganic PCMs are mostly salt hydrates that tend to have high thermal conductivities and latent heat capacities; however, they are generally corrosive, leaving them unsuitable for many applications. Also, another issue is supercooling; nucleation occurs at a low rate at the fusion temperature, so supercooling is needed to achieve an acceptable rate [94]. The focus of this paper is on organic PCMs, however quality information about the properties of inorganic PCMs can be read about in the published research [95,96]; these papers exhibit some methods to increase the thermal properties and stability of inorganic PCMs.

4.1.2. Organic PCMs

Most organic PCMs have high latent heat capacities, are non-toxic, non-corrosive and inert; however, they do not exhibit high thermal conductivities [94]. Organic PCMs are a broad material group and encompass fatty acids, esters, paraffins, alcohols and other organic compounds, which have been explored in detailed depth previously [33]. Typically organic PCMs have the benefit of being relatively low cost which have been explored well in some of the previous papers [9,21]. Since the thermal conductivities of organic PCMs are poor, they must be adapted to respond to rapid thermal energy discharges to be of use. There are methods to enhance the intrinsic thermal conductivity of a polymer via an ionization, which stiffens and extends polymer chains, creating improving packing of the molecules thus increasing the thermal conductivity [97,98].

Paraffin waxes are organic PCMs, comprised of n-alkanes, that can have phase change temperatures in the range −20 °C – 200 °C

Table 2
Thermal properties of some different phase change materials [86].

Materials	Melting temperature(°C)	Latent heat(kJ/kg)	Reference
Salt hydrates			
Na2SO4.10H2O	32.4	254	[87]
Mn(NO3)2.6H2O	26	125.9	[8]
CaCl2.6H2O	29	187	[88]
MgCl2.6H2O	117.2	168.6	[8]
AlCl3.6H2O	192	280	[89]
Organic			
Paraffin (C16–C18)	21.2	152	[8]
PEG (600 g mol– 1)	21.6/27.7	106.5	[90]
Stearic acid	54–56	186.5	[43]
Palmitic acid (97%)	61	203.4	[44]
Lauric acid	40–43	167–171	[91]
Butyl stearate	19	200	[92]
Dimethyl sebacate	21	135	[92]
Eutectics			
Inorganic–inorganic			
66.6% CaCl2.6H2O + 33.3% MgCl2.6H2O	25.2	127	[8]
58.7% Mg(NO3)2.6H2O + 41.3% MgCl2.6H2O	59.2	132.2	[8]
Organic–organic			
Capric–lauric acid	18	140.8	[88]
Lauric–palmitic acid	32.7	145	[93]
Lauric–stearic acid	34	150	[93]
Palmitic–stearic acid	51	160	[93]
Organic–nonorganic			
66.6% Urea + 33.4% NH4Br	76.2	161	[8]

Table 3
Different dimensions/types of carbon nanofillers.

Dimensionality	Allotrope
0D	Fullerene
0D	Onion-like carbons
1D	Nanofibers
1D	Carbon nanotubes
2D	Graphene/graphite
3D	Graphene/graphite-aerogel/array/network
3D	Expanded graphite/activated carbon

[7] that spans values needed by many applications, such as electronics cooling and domestic heating. The phase change temperature may be adjusted by altering the *n*-alkane composition of the wax, with larger values of *n* resulting in larger melting temperatures and higher latent heat of fusion. Crystallization of the n-alkanes releases large amounts of latent heat [7]. The issue with paraffin wax is that its thermal conductivity is poor; to improve this important aspect of the PCM there have numerous attempts, which have been explored in later sections. Improving crystallization is desired. Crystal growth can be controlled in polymers, via temperature and time control [99] which could prove beneficial to PCM properties.

Eutectic PCMs are mixtures of two or more organic PCMs that creates differences in the melting temperatures and heat capacities. They are employed to sharpen the temperature range of which the melting occurs in order to better suit a specific application [7].

5. Carbon nanofillers

Carbon can exist in many different morphologies, giving rise to novel properties and different benefits due to shape. For examples, some allotropes of carbon are presented in **Table 3**. Graphene is an allotrope of carbon, it is 2-dimensional and consists of a lattice of hexagonally arranged carbon atoms. Graphene can also be described as single layer graphite and shares much of the same thermal properties.

Suspended graphene has one of the highest thermal conductivities of all materials and in the in-plane direction it is in the range of 2000–4000 $Wm^{-1}K^{-1}$ [19]. The largest possible thermal conductivities are acquired with isotopically pure graphene with large grain sizes [19]. Imperfections in the graphene structure or residue from its manufacture will reduce the thermal conductivity by increasing the amount of phonon scattering.

The thermal resistance of multi-layer graphene samples does not significantly increase when compared to single layer graphene sheets [20]. Therefore, it is possible that the thermal resistance is dominated by the interfacial resistance between graphene and the material it is interfaced with [19]. Phonon propagation is significantly hindered by the boundary scattering [21]. However, experimental studies using few layer graphene (FLG) suggest that as the number of layers of graphene increase, the thermal conductivity decreases. When the number of layers of graphene exceeds four the thermal conductivity will tend to that of graphite [61].

It is expected that different morphologies of carbon will have different thermal properties and will interact with PCM in differing ways, therefore it should be known which types of carbon provide the best thermal enhancements, amongst other relevant properties, when composited with PCMs.

The filling particle chosen will depend on many factors such as TCE and enthalpy retained, however the cost of producing/purchasing different carbon nanomaterials can vary due to their production method and will likely be an important determining factor. The more pristine a carbon nanomaterial is, the higher the thermal conductivity, and generally, the higher the production cost is. Graphene of a large area and uniform morphology production is a high cost process [100]. The cost of carbon nanomaterials is likely to decline due to their high demand for many varied applications [101] and efforts being made to improve their production as well. For example, in the year 2007 the cost of multi-walled carbon nanotubes was $200/g [102] and as of now, 2021, the price is $0.90/g [103]; this trend is likely to continue.

The columns labelled TCE (thermal conductivity enhancement) in Table 4 describes the increase in thermal conductivity, relative to the matrix material of the composite; it is defined as $100\times(K_c - K_m)/K_m$, where K_c and K_m are the thermal conductivity of the composite and the pure matrix, respectively [104]; these values are reported for the thermal conductivity of the composite in its solid state. The TCE value is useful to understand the relative improvement in conductivity, resulting from the addition

Table 4
Properties of CPCMs with 3D carbon additives.

3D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/ Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
CNT array	Array height: 50–1000 μm Radius: 5 nm	RT100	–	5134%	12.3	195.3/192.8	84.8%	Melt infiltration	2019 [105]
Expanded graphite	–	RT44HC	35 wt%	2900%	6	152.5/-	67.4%	Melt mixing	2015 [106]
Expanded graphite	–	Paraffin	10 wt%	1160%	3.83	Roughly 118/-	–	Impreg with roll mixer	2010 [107]
Graphite foam	–	Paraffin	48.8 wt%	646%	1.568	220/-	126.9%	Molten vacuum impreg	2015 [108]
CNT sponge	Pore size: ~300 nm - 1 μm	Paraffin	20 wt%	500%	1.2	Roughly 120	88.2%	CVD, solution infiltrated	2012 [109]
Graphite foam	–	Erythritol	35 wt%	423.6%	3.77	251.2/-	75.8%	Incipient wetness impreg	2015 [110]
rGO/CNT aerogel	–	Paraffin	0.876 and 0.438 wt%, respectively	318%	0.836	245.5/ -	92.6%	In situ self-assembly, vacuum impreg.	2019 [111]
Carbon foam	–	Paraffin	–	238%	1.198	164.98/-	87%	Vacuum impreg.	2013 [112]
Graphene framework	Pore sizes: 0.7–8 nm	Dodecanoic acid	40 wt%	159.1%	0.35	114.5/110.8	58.9%	Solution impreg.	2020 [113]
Carbon aerogel	Interlayer distance: 50–200 μm	1-hexadecanol	2 wt%	107.9%	0.53	220/-	92%	Chitosan-derived, infiltrated	2017 [114]
Graphene oxide network	–	PEG6000	0.5 wt%	63%	0.344	218.9/213.2	98.7/97.9%	Ca^{2+} cross-linked, sonication.	2019 [115]
Carbon powder	–	PEG	2.5 wt%	45%	0.29	–	–	Melt stirring	2019 [116]

of a filler, relative to other fillers and irrespective of PCM thermal conductivities.

5.1. 3D carbon nanofillers

A 3D nanomaterial inclusion has all dimension lengths > 100 nm or can be groups of lower dimensional materials (1D, 2D or 0D) which are chemically bound or interact to form a network/structure with all dimensions > 100 nm. The difference between a 3D nanomaterial and a bulk material is the relative surface area and properties which are imparted by the small scale of the nanomaterial; 3D nanomaterials tend to have features that are on the nanoscale (pores, network branch dimensions) that have useful qualities. Table 4 displays the properties that different 3D carbon nanomaterials impart on various organic PCMs and is ordered in decending TCE.

Qui et al. [115] prepared a 3D graphene oxide network crosslinked with Ca^{2+} and composited it with the PCM - 'PEG6000'. Graphene oxide was made from natural graphite flakes using a modified Hummers method, Ca^{2+} crosslinks were formed and a gel that was then vacuum freeze dried to obtain the 3D network. Different amounts of PEG6000 were added to the 3D network in situ and then mixed with an ultrasonic disperser. **Fig. 9** shows the morphology and the preparation steps of the network. It was found that the thermal conductivity of the PEG/GO network CPCM could be improved by 63.0–87.7% for a range of GO fractions 0.5–6 wt%. The Ca^{2+} allows maximisation of the PEG content so that the melting latent heat may be maximised, as an increased fraction of filler has the effect of reducing the heat capacity.

Cao et al. [111] prepared a rGO/CNT aerogel from a dispersion of GO and CNTs, which self-assemble into an aerogel. The aerogel was obtained by creating a hydrogel of rGO/CNT which self-assembled in an oven for 5 h at 95 °C. The aerogel is the vacuum impregnated with paraffin.

Fig. 10 shows a schematic with the preparation steps of an aerogel. The best enhancement of thermal conductivity was obtained from using a 1:0.5 ratio of rGO:CNT, giving a 318% increase. Different ratios will give different thermal properties; the thermal conductivity can be regulated by the adjustment of the ratio. The high increase in thermal conductivity from the 1:0.5 ratio can be attributed to the functional groups imparted on the aerogel by the rGO, which enhance the interfacial reaction between it and the paraffin matrix. Also, the effectiveness of the vacuum impregnation method creates a small loss of enthalpy change due to the reduction of void space within the aerogel.

Xiao et al. [112] synthesized and thermally characterized a paraffin/carbon foam CPCM. The CPCM was made via vacuum impregnation of the foam with molten paraffin. A TCE of 238% and enthalpy retainment of 87% was reported, however the authors did not state the loading fraction for these values. Their results are congruent with other literature [109,115], but further study must be done to understand the required fractions.

Fig. 11 shows the pore structure that is to be infiltrated with paraffin wax, with a pore size of about 1.5 mm. It is believed that as the porosity increases the thermal conductivity decreases. Xiao et al. concluded that vacuum impregnation is superior to without; an impregnation ratio of ~96% with vacuum and ~92% without is reported.

Fang et al. [114] fabricated a carbon aerogel/1-hexadecanol CPCM using a biologically derived chitosan powder. The carbon aerogel was then infiltrated with the 1-hexadecanol PCM. Thermal conductivity enhancements of 107.9% were reported for a filler fraction of 2 wt% carbon aerogel. The carbon aerogel provides a high porosity for the PCM to infiltrate, whilst increasing the contact with and thus thermal conductivity to the high thermal capacity PCM. High porosity also aids in the shape stabilisation of the CPCM, with the carbon aerogel reported to be able to hold 50 times its weight in PCM; capillary forces are at work here holding the PCM in the interlayers of the carbon aerogel. They state that the shape was able to be maintained at 70 °C for more than 30 min.

Zhang et al. [105] made a RT100/CNT array CPMC via infiltrating the CNT array with melted PCMs. The array was comprised of aligned, unidirectional MWCNTs with 3–7 walls per CNT and a

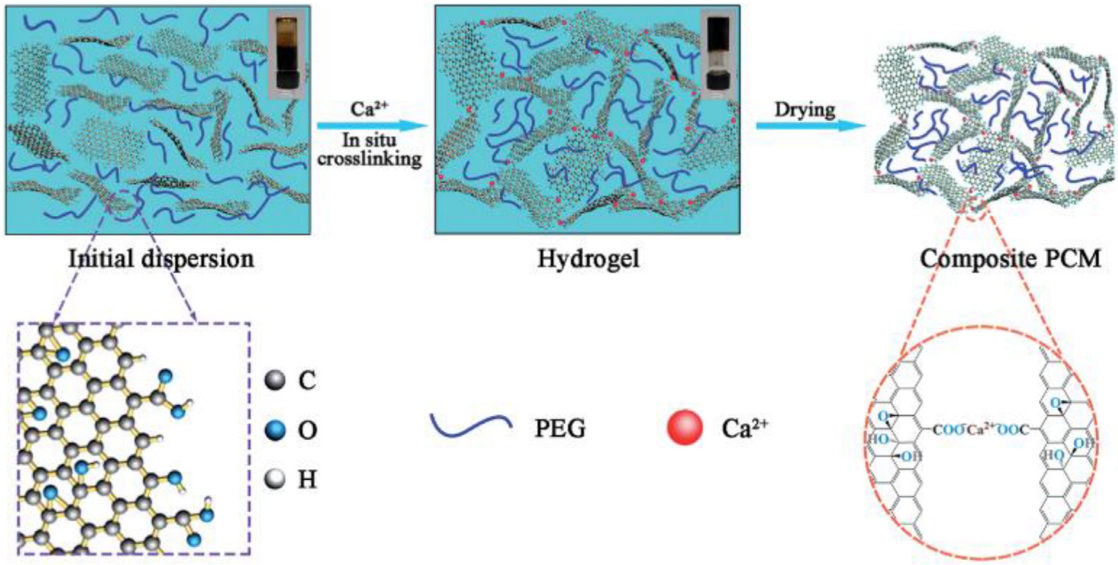

Fig. 9. A diagram illustrating the preparation of a GO network with in situ PEG filling [115].

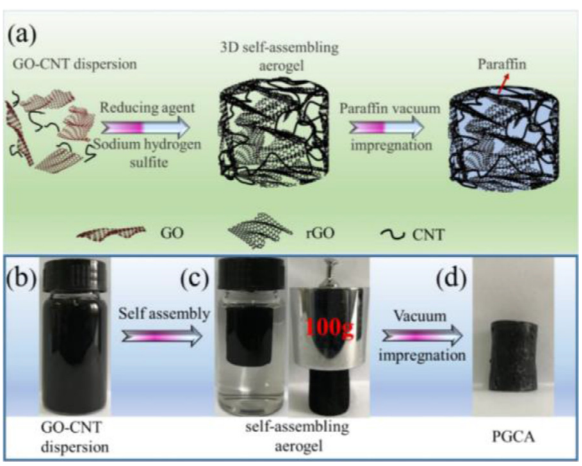

Fig. 10. (a) Schematic diagram indicating the steps of preparation of PGCA SSPCMs; (b) A photograph of rGO/CNT dispersion; (c) A self-assembled aerogel and (d) PGCA SSPCMs [111].

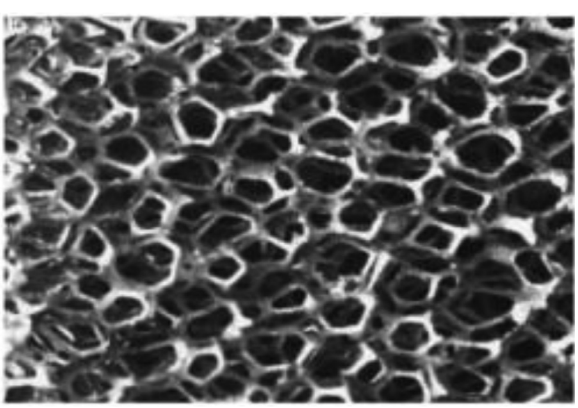

Fig. 11. An optical image of porous carbon foam [112].

porosity of 99%. Their results show at 52 times increases in the thermal conductivity of the CPCM in the Z direction (parallel to MWCNT axis) compared to the pure PCM; the authors did not share the ratio of CNT array to RT100 they used, however with a retention of 84.8% of the enthalpy, this suggests that the majority of the composite was RT100 (rule of mixtures) and this value being similar to Chen et al. [109] would suggest it is lower than 20 wt%. There was also a thermal conductivity enhancement of 17.7 times in the X-Y direction. However, this CPCM has an anisotropic nature and hence its application will be limited. 200 reverse cycling tests were also performed on the composite with minimal on the melting and freezing enthalpy, 0.1% and 0.2% loss respectively. SEM characterisation was performed after the 200th cycle, which showed that the morphology remained very similar which credits the shaped stability and its capacity to not leak after cycling. This ability is engendered by the porosity of the array and the capillary forces it exerts on the PCM to retain it [52,82].

Chen et al. [109] fabricated a paraffin infiltrated CNT sponge. 3D CNT sponges were produced by chemical vapour deposition; MWCNTs of 20–30 nm diameter and $\sim$10 μm length were overlapped and self-assembled into an isotropic porous network. The paraffin was solution infiltrated into the sponge to form the composite. A thermal conductivity enhancement of 600% can be achieved for a loading of 20%. Below loading fractions of 87% paraffin the enthalpy value is lower than the inherent value of paraffin. Despite the rule of mixtures, the composite experiences an increase in the enthalpy above a loading of 91% paraffin. X-ray diffraction performed by Chen et al. suggested that the existence of intermolecular C–H—-π interactions, which could apply compressive stress to the matrix during a solid-liquid transition and aid in increasing the enthalpy of the CPCM. Using the methods proposed in this study and by modifying the loading fraction the thermal properties of CPCMs can be adjusted.

Singh et al. [116] added 10- 50 µm sized carbon powder prepared by granulating carbon plates with a taper drill at 0.78 and 2.5 wt% to polyethylene glycol (PEG) which was mixed in a beaker in PEG molten state. Increases in thermal conductivity were reported at 22.7 and 31.8%, respectively. The effectiveness of the preparation method in achieving a dispersed filler is questionable because carbon particles of 10–50 µm will agglomerate [117], not disperse, and will not reflect the potential of the filler to enhance the thermal conductivity of the composite.

Ling et al. [106] explored the addition of expanded graphite to Rubitherm's PCM (RT44HC) and found that an increase in thermal conductivity by a factor of 20–60 was possible. The thermal conductivity increases with increasing packing density and weight percent of expanded graphite. The effect of packing density was explored by compressing the composites to different densities; at a density of 300 kgm^{-3} a thermal conductivity of 4.3 $Wm^{-1}K^{-1}$ was recorded, and for 900 kgm^{-3} it was 10.7 $Wm^{-1}K^{-1}$. These value increases can be explained by the graphite network morphology being compacted, shortening the thermal conductivity path length. The authors believe that their results can be translated to any type of PCM/poor thermal conductivity material and that expanded graphite can improve them. Expanded graphite has porosity in its structure and so will likely alter the effect of convective thermal transport within the material and increase the contact interface of the graphite with the matrix. Using expanded graphite in higher quantities for industrial purposes would be suitable, due to the low cost of the material and its production [118].

Dimberu et al. [113] fabricated a graphene framework/dodecanoic acid CPCM via solution impregnation. The framework was produced via a pyrolysis method using Na_2CO_3 listed in the paper. The resulting, shape stable, network was solution impregnated with dodecanoic acid, creating the CPCM. The authors characterised several loadings (20, 30 and 40 wt%). The thermal conductivity shown in **Table 4** is much lower than for other CPCMs with a similar loading, this could be explained by impurities left behind by the Na_2CO_3, or defects introduced to the graphene by pyrolysis that would act as phonon scattering centres.

Activated carbon is a filler of interest due to the versatility of preparation methods and precursors available; materials such as food waste (coconut shells, fruit stones, waste from coffee production), polymeric raw materials, and many other types of waste can be used to produce activated carbon [119]. It has a high surface area and a porosity, which can lead to a superior leakage resistance of the PCM [120] and it can take on many morphologies, dependent on precursor and production. Hussein et al. [121] prepared activated carbon nanosheets using a hemp precursor and via sonication, then it was mixed with a eutectic blend of oleic and capric acid. Thermal conductivity enhancements of 12%, 32%, 35%, 45% and 55% were observed for 0.02, 0.04, 0.06, 0.08 and 0.1 wt% additions of activated carbon, respectively. These results are relatively good when compared to others; for example the 63% TCE at 05.wt% for a graphene oxide network by Qui et al. [115]. The activated carbon composite lost latent heat as the wt% of activated carbon increased, hence, the crystallinity of the PCM is unlikely to be significantly improved.

5.1.1. Functionalisation of 3D nanofillers

Surface functionalization may increase the interfacial thermal resistance [81] but could also have the effect of reducing thermal resistance [127]. Physical and chemical interactions between the nano-filler and PCM material must be considered in all types of CPCM enhancements as the viscosity/convective heat transport, thermal conductivity, thermal stability, cycling stability and enthalpy will be affected. Table 5 displays the properties of various CPCMs made with different functionalized 3D carbon nanomaterials and is ordered in decending TCE.

Yang et al. [123] fabricated a cellulose/GNP aerogel with 3D network structure, that was then infiltrated with polyethylene glycol (PEG). The cellulose and GNP fraction used was 5.5 and 5.3 wt% respectively. The graphene nanoplatelets were annealed to remove oxygen groups and defects, thus enhancing its thermal conductivity by the removal of phonon scattering centres. Hydrogen bonding between the cellulose and PEG matrix increased the forces and thus encapsulating effect on the PCM, enhancing the leakage properties at elevated temperatures. After 50 thermal cycles changes in the enthalpy values were minor. This shows that 2D fillers such as GNPs may be functionalised with molecules to form an interconnected 3D network, enhancing the thermal transport, leakage resistance and thermal stability.

Although Shao et al. [124] prepared a polymer (polyamide-6)-BGF composite, the results are useful for the study of PCM enhancement. If the PCM used has the same degree of intermolecular interaction with the porous network, then thermal conductivity enhancements will be analogous. The boron nitride nanosheets (BNNS) and graphene were bonded by a condensation reaction of the graphene's oxygen groups and the functional groups of the BNNS; this created a 3D network structure that was then grafted with the polyamide-6 (PA6) molecules, improving dispersion and matrix interactions.

Wang et al. [125] attempted to functionalize boron nitride nanotubes (BNNTs) with rGO to form a functionalized network however, SEM imaging showed that the morphology took on the characteristics of pure rGO, at all fractions of BNNTs. At a 1:1 wt ratio of BNNT to rGO the BNNT can be seen to form interdigitated network structures, either on the surface of rGO or wrapped in its layers. The composite benefitted from an increased enthalpy of fusion, which the authors attribute to the BNNT's effect of hindering melting and crystallization of PEG through intermolecular attraction; it is possible this is the case, but it may be that the intermolecular attraction provides a site for enhanced crystallization of the PEG like in other studies [14]. Compression testing showed that the composite could withstand 75% strains for 10 cycles and retain its original dimensions, this would make the composite useful for certain applications. The composite can also withstand temperatures of 100 °C for at least 10 min with almost no leakage, proving its shape stability.

Cao et al. [126] prepared a CPCM from octadecylamine-functionalized rGO (C_{18}-rGO) and polynorbornenes which self-assemble into a 3D network structure. Thermal stability and cycling tests were conducted on it; after 100 cycles the melting enthalpy had dropped by 8.7% while it was a 6.6% drop after 200 cycles when compared to the composite that has not undergone cycling. These values prove that the functionalized network structure has a good ability to retain its enthalpy after cycling, either by retention of the paraffin PCM or by retaining the crystalline structure of the PCM. When the CPCM was heated to above the melting temperature of paraffin there was no leakage in the 5 wt% composite; in the lower wt% paraffin was seen to leak out onto the filter paper they were upon. Unfortunately, the authors did not report thermal conductivity values, but did report that there was an improvement when compared to the pure PCM; it would be useful to compare values with similar unfunctionalized rGO composites. Similar work has been done by Li et al. [128] who reported a loss in melting enthalpy of 2.2% after 500 cycles. Optimising the interactions of the filler with the matrix can prolong the lifetime of CPCMs.

5.1.2. Encapsulation of 3D nanofillers

There has not been much research into the encapsulation of 3D carbon nanofilled PCMs, or that the research papers are low in quantity and hard to find. This may be due to the fact that adding 3D nanostructures to a PCM has a similar effect to encapsulation by improving thermal and cycle stability, therefore having no useful effect.

5.1.3. Summary of the use of 3D carbon nanofillers with PCMs

All the studies summarised in **Tables 4** and **5** are presented in graphical format in **Fig. 12** to improve comparison and understanding. Regardless of the dimensions of the 3D networks, there does not seem to be any thermally percolating behaviour; this, mostly likely, is due to the fact that the 3D fillers are already connected in a network like manner (as was presented in **Fig. 11**) and higher values of wt% will only increase the amount of high thermal conductivity material in the composite. There are not many studies that explore the use of 3D carbon additives at high wt%, therefore

Table 5
Properties of differently functionalised 3D carbon nanofilled PCM composites.

3D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
Graphene aerogel	Pore size: 3–10 μm	1-Octadecanol	11.4% GNP and 0.6 wt% rGO	2474%	5.92	202.8/250.6	85.1%/81.8%	Vacuum impregnation	2016 [122]
Cellulose/ GNP	–	PEG	5.5 wt% cellulose and 5.3 wt% GNP	462.5%	1.35	156.1/148.9	87.3%/88.0%	Vacuum impregnation	2016 [123]
BNNS/ Graphene	–	N/A	1.6 wt% BNNS, 6.8 wt% graphene	356%	0.891	–	–	BGF prepared by hydro-thermal reduction	2016 [124]
BNNT/rGO	Pore size: 200 μm	PEG	1.5 wt%	48.3%	0.43	195.6/-	103.8%/-	Vacuum infiltration	2019 [125]
C_{18}-rGO	–	Paraffin	5 wt%	–	–	144.2/144.6	72.4/71.1%	Solution mixing followed by reflux	2020 [126]

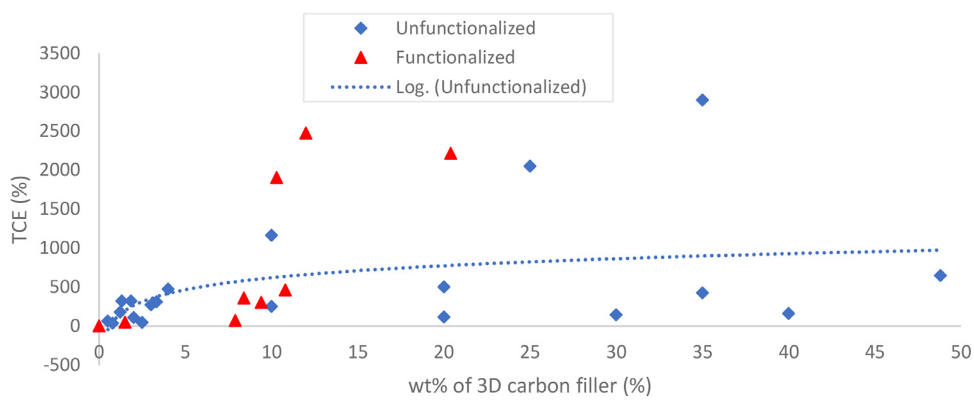

Fig. 12. TCE vs wt% of 3D carbon filler for both functionalized and unfunctionalized fillers.

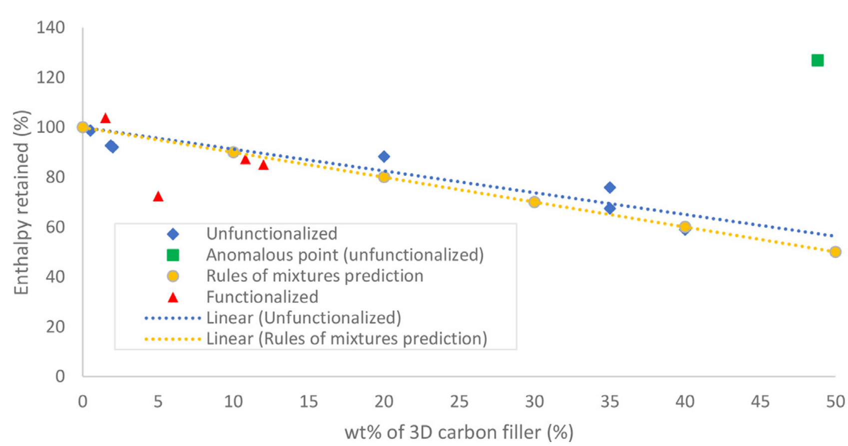

Fig. 13. Graph of composite enthalpy retained vs wt% for the types of 3D carbon fillers.

the data presented in Fig. 12 is limited and may not fully represent the behaviour. Considering the above, it is apparent that the thermal conductivity increases at a low wt% is more rapid when compared to the increase at higher values of wt%. The thermal conductivity does show an increasing trend as wt% increases.

Fig. 12 appears as if there is percolation behaviour from the functionalized 3D carbon fillers, occurring at around 10wt%, however this may be due to the limited data available from current articles; it is unlikely there is thermal percolation as the structures are already connected networks.

In the meantime, Fig. 13 demonstrates the effect that adding filling particles without latent heat of phase change properties has on the latent heat of the base PCM; mostly they detract from the desired property. In exceptional circumstances, such as the anomalous point, the filler particles aid in forming a crystallized PCM, resulting in higher latent heats. The experimental data from the literature closely follows that predicted by the rule of mixtures. Graphed in Fig. 13 are the changes in composite enthalpy for functionalized vs unfunctionalized and the rule of mixtures. It appears as though the functionalized additive filled PCMs would obey the rule of mixtures if there was more data, when ignoring the value at 5 wt%; this point deviates possibly because of the quantity of functionalizing molecule polynorbornene added.

Using 3D carbon nanofillers (or assemblies of lower dimensional materials) in CPCMs can impart useful qualities on the composite such as low enthalpy losses and low leakage after cycling, shape stability, and thermal stability. Such qualities are prevalent in 3D nanostructure filled PCMs due to the pores of the networks and the capillary forces they exert on the PCMs. Another advantage of some 3D nanomaterial CPCMs is that they offer resistance

Table 6
Properties of 2D carbon nanofilled PCM composites.

2D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/ Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
Graphene	Thickness: 1 nm Planar dimensions: 10 μm	Paraffin	20 wt%	17,900%	45	–	–	Melt shear mixing	2014 [129]
Expanded graphite	–	Stearic acid	25 wt%	12,828%	23.27	154.0/-	75%	Molten impregnation	2016 [130]
EG	–	RT44HC	35 wt%	7900%	16.0	152.5/-	67.4%	Melt mixing	2015 [106]
Graphene	Thickness: 15 nm Length: 25 μm	Paraffin	38.5 wt%	2800%	7.25	314.53/309.41	113%/113%	Ultrasonication	2014 [131]
Annealed graphene	–	1-octadecanol	10 wt%	1514%	3.55	195/-	78%	Solution sonicated	2014 [104]
Expanded graphite	–	Stearic acid/benzamide	16 wt%	1485%	4.177	176.03/174.02	88%	Melt stirred	2019 [132]
Graphene	Thickness: < 2 nm Planar dimensions: < 2 μm	Beeswax	0.3 wt%	1056%	2.89	190.77/-	131%	Molten sonication	2017 [133]
GNP	12 ± 2 nm	Hexadecane	0.0075 wt%	121%	0.5525	–	–	Sonication and stirring	2018 [81]
GNP	Thickness: 6.5–9 nm Planar dimensions: 1.5–2 μm	Palmitic acid and DMF	4.9 wt%	48.8%	0.224	–	–	Molten sonication	2019 [134]

Table 7
Density changes of paraffin loaded with different nanoparticles, all at 20% [119].

Material	$P_{mixture}$ (kg/m^3)	$\rho_{paraffin}$ (kg/m^3)	Changes in $\rho_{paraffin}$ vs base Paraffin value (%)
Base paraffin	–	880	
Graphene - Paraffin	1280	1039	18.0
MWCNT - Paraffin	1081	845	4.0
Al - Paraffin	1235	874	0.6
TiO_2 - Paraffin	1233	881	0.1

to strain and will return to their original shape. There are many ways to synthesise 3D nanostructures, including starting from 1D and 2D nanomaterials which allows a wide range of precursors and methods to be used, which gives varied properties. A disadvantage is that the liquid state thermal conductivity of the composites is likely low due to the hindered convective thermal transport, which is the dominant mechanism.

5.2. 2D carbon nanofillers

Table 6 displays the properties of various organic PCMs enhanced with different 2D carbon nanomaterials and is ordered in decending TCE. Graphene-PCM composites are able to enhance the phase change enthalpy values when compared to the pure PCMs [131,133] which is not typical of CPCMs. Warzoha et al. [131] managed to increase the enthalpy by 13% for a large loading of 38.5 wt% (or 20% volume) of graphene. This is highly unusual due to the fact that a larger wt% of filler will be replacing the high heat capacity PCM, as per the prediction from the rule of mixtures; however density measurements were performed, which show that there was only a 2% reduction of paraffin in the composite. This suggests that the paraffin crystallinity has been increased by the presence of the graphene; crystalline paraffin is denser, and this theory would also support the larger phase change enthalpy as crystalline paraffin has a higher phase change enthalpy. **Table 7** demonstrates that different types of filler will influence the density of the paraffin, where ρ is the density value. Since cycling tests were not performed it is not clear whether these results will transfer to practical applications and retain crystallinity after subsequent cycles, therefore further research should be recommended in this area.

Goli et al. [129] also demonstrated the ability to use graphene at high wt% loadings whilst preserving the latent heat of the composites. The high TCE achieved by Goli et al. [129] can be attributed to the coupling interaction between the paraffin molecules and the graphene, which lowers the interfacial resistance, improving phonon transport; it is likely the matrix-filler coupling because a similar massive increase in TCE of 5900% occurs with 1 wt% graphene (from the same study), which is not likely to form a thermally percolating network at such a low filling ratio. Why this graphene-paraffin CPCM performs much better than the others of similar composition is not yet well understood.

Guoqing et al. [104] found that by using thermally annealed graphene that is defect free, a PCM composite can be made which has a 6 fold increase in thermal conductivity when compared to the same composite with pristine graphene, and a 16 times increase when compared to the PCM alone. The thermally annealed, 2D, defect free graphene sheets were made by removal of defects and oxygen functional groups; this material was used at 10 wt% and gave a thermal conductivity of 3.55 W/m K. The structure of the annealed graphene removes phonon scattering centres, increasing the thermal transport; this explains the 6-fold increase in thermal conductivity when compared to the pristine graphene CPCM. At the moment the scalability of production of annealed pristine graphene is not known, however it is unlikely to be suitable for large scale production due to the steps taken in the preparation method [104].

Amin et al. [133] experimented with graphene/beeswax CPCMs and found that at a 0.3 wt% loading of graphene the thermal conductivity of the PCM could be enhanced by 22.5% to a value of 2.8 W/mK. A benefit of using beeswax is that it is a natural and renewable PCM. The molten beeswax and graphene were combined and sonicated to achieve a uniform dispersion. Characterisation of the CPCM was performed using FTIR to understand the interactions between the two materials; the resulting peaks from FTIR showed no significant unexpected peaks, meaning that no chem-

Table 8
Properties of PCMs filled with functionalized 2D fillers.

2D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/ Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
GO	Pore size: 8 nm	Paraffin	55.19 wt%	360%	1.32	59.12/60.73	44.8%/45.9%	Vacuum impregnation	2013 [31]
rGO–Co	–	PEG	20 wt%	159.0%	0.523	–	–	Mixing/melt vacuum impregnation	2020 [136]
GNP	12 ± 2 nm	Hexadecane	0.01 wt%	111.4%	0.526	–	–	Sonication and stirring	2018 [81]
GO-ADCT-NH_2	–	PEG	9.6 wt%	111%	0.983	157.6/141.8	85.6%/81.3%	Heated, prolonged stirring	2017 [137]
Urethane-GO	–	PUB	0.5 wt%	61.5%	0.302	–	–	Ultrasonication	2018 [138]
OA-rGO	–	PA	2 wt%	50.2%	0.419	181.9/189.3	89.6/88.4%	Heated stirring	2015 [139]
PDA-rGO	–	EmPEG	3 wt%	33.3%	0.52	160.6/158.3	89.3/88%	Sonication and stirring	2020 [140]
Fe_3O_4-GNS	Thickness: 3.72 nm	PCM polymer	3.85 wt%	27.8%	0.317	100.1/99.3	99.4/99.7%	Solution sonication	2017 [141]

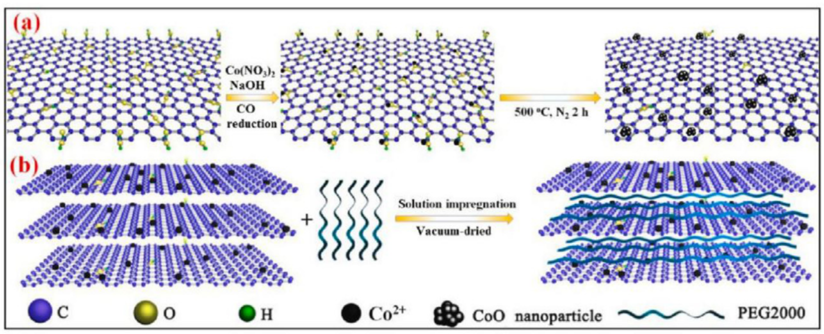

Fig. 14. (a) the synthesis of rGO-Co and (b) its composite with PEG [136].

ical bonds were formed between them. Contrary to other publications summarised in **Table 6**, the composite experienced an increase in the latent heat of solidification of 31% for a loading of 0.3 wt% graphene. The authors report that the latent heat has an almost linear relationship of increasing latent heat to increasing mass fraction. Studies have shown that micro-scale inclusion in a PCM matrix may improve heat storage, but as particle diameter decreases the heat storage capacity decreases due to Brownian motion [135].

Ma et al. [132] made a eutectic PCM from a mixture of stearic acid and benzamide analysing the ratios for which the peak enthalpy is achieved; additions of expanded graphite were added at different loadings to a PCM mixture and stirred for 1 h above the melting point of the PCM, allowed to cool to room temperature and solidify, and then were compressed into disks for characterisation. Leakage tests were also performed to characterise the effectiveness of different wt% at combating this; the lowest mass loss ratio was obtained for 12 wt% loading. As the loading fraction increases the enthalpy value decreased, as expected, also the thermal conductivity increases. Thermal cycling testing was also performed; 100 thermal cycles were applied to the eutectic PCM mixture alone and the mixture with 12 wt% expanded graphite addition. The results shown in their table and figures for the 12 wt% CPCM contradict, with data in the table showing a 2% increase in enthalpies by the 100th cycle however they conclude that they have a loss of 2% after the cycling. Either result is satisfactory; with a loss of 2% enthalpy at the 100th, this shows that leakage is minimal and the PCM is contained proving good thermal reliability.

Wu et al. [130] created a CPCM via impregnating an expanded graphite (EG) filler with melted stearic acid followed by its compression into a block, achieving a massive TCE of 129.3 times. This result is very good however the CPCM exhibits anisotropic properties in radial and axial direction with the axial direction being up to 4 times lower in thermal conductivity with increasing packing densities; the explanation may be that increased packing density forces the 2D flakes in close proximity, and alignment parallel to the compression face, creating a more connected, yet anisotropic microstructure. Graphite has lower thermal conductivity perpendicular to the 2D plane, between layers of graphene supplementing this explanation [56]. The CPCM has an acceptable value of latent heat, and it is noted that the latent heat decreases linearly with additions of higher wt% of EG. Thermal stability testing was performed, showing that a higher wt% of EG causes leakage of stearic acid at a lower temperature and also that the onset of leakage occurs at roughly 170 °C, ending at 227 °C when all the stearic acid evaporates. Cycling stability was also carried out and showed that lower packing densities and a larger wt% of EG improve the stability, resulting in less stearic acid loss.

Mishra et al. [81] studied the thermal conductivity of graphene nanoplatelets (GNP) at different loadings (0.001, 0.0025, 0.005, 0.0075 and 0.01 wt%) and at different states of the CPCM. At liquid, liquid-solid transition and solid states, the thermal conductivity enhancements for 0.0075 wt% were 7.9%, 280% and 121.4%, respectively. The TCE in the liquid state was the lowest and increased up until 0.0075 wt%, falling at higher fractions; this behaviour could also be seen in the solid state but with larger TCEs and the behaviour was reversed in the liquid-solid transition phase, showing a decrease in TCE with increasing wt% up until 0.0075 wt% where higher fractions improved the TCE. This could be explained by increasing agglomeration of the GNPs at higher wt%

Table 9
Properties of differently encapsulated 2D carbon nanofilled PCM composites.

2D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W $m^{-1}K^{-1}$)	Melting/ Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
Graphite oxide and GNP	Diameter: 100–200 μm	Paraffin	2 wt%	260%	0.90	187.10/184.30	97.4/97.2%	Pickering emulsion method	2020 [145]
Graphene	Diameter: 27.4 μm	Stearic acid	5 wt%	63.4%	0.335	181.6/175.3	93.0%/90.0%	Agitation, filtration, and vacuum drying	2015 [146]
GNS	–	1-dodecanol	1 wt%	53.3%	0.46	-/206.6	-/95.4%	Sonication	2016 [23]

for liquid and solids states, but for the liquid-solid transition it is unclear, thus further research must be carried out to extend the understanding.

5.2.1. Functionalisation of 2D nanofillers

Table 8 displays the properties of various organic PCMs loaded with different functionalized 2 carbon nanomaterials and is ordered in decending TCE. Graphene oxide can be considered a functionalized 2D allotrope of carbon due to its carboxyl, hydroxy and epoxide groups [142] which will enhance matrix interaction and dispersion within a PCM of the same polarity. Mehrali et al. [31] studied the thermal conductivity, stability and latent heat changes for a graphene oxide-paraffin CPCM with a large filling ratio of GO. Thermal degradation of the paraffin wax was delayed by the GO due to its higher degradation temperature and the morphology of the GO walls, which prevent the paraffin from being diffused away. After 2500 thermal cycles only a 1.71% reduction in melting enthalpy was seen, which is a testament to the effectiveness of this composite in preventing PCM leakage. Although the thermal conductivity enhancement was improved for this CPCM, it was made at the detriment of the enthalpy values which have reduced to about 45% the original value; this is due to the reduced amount of PCM in the composite.

Composites with good solar to thermal absorption properties can be made by methods detailed by Li et al. [136]. Reduced graphene functionalized with cobalt (Co) were also decorated/embedded with CoO nanoparticles of varying amounts. **Fig. 14** shows the synthesis and morphology of the resulting composite and it can be seen that the CoO are embedded in the surface of the rGO, which aids in its dispersion. The composites were all prepared with 80 wt% of PEG2000 but with varying amounts of rGO and CoO. Solar-to-thermal light absorption characteristics increased with larger proportions of CoO, given that the process depends on the thermal conductivity of the composite, this means that increased CoO content improved thermal conductivity. It is likely that CoO aids in creating a more percolated network, improving the thermal transfer pathways.

Polyurethane (PU) can be considered a phase change material [143] and its TCE has been improved with urethane-functionalized graphene oxide by the work of Du et al. [138]. Graphene oxide was covalently functionalized with urethane molecules and subsequently mixed into a waterborne PU binder (PUB) matrix. The thermal conductivity enhancements of 61.5% and 29.9% were achieved for 0.5 wt% and 1 wt% respectively. Normally, it is expected that the higher weight fraction would have a larger TCE effect, however it is likely that the functionalization did not improve the dispersion quality at higher concentrations and caused particle agglomeration.

Tang [141] used Fe_3O_4 functionalized GNS (graphene nanosheets) to enhance the thermal properties of a synthesised, form-stable, PCM whose creation is detailed in the article. The benefit of using a Fe_3O_4-GNS additive is that it allows for the conversion of magnetic fields and solar energy to thermal energy; this composite would therefore have applications in novel energy harvesting methods. From the CPCM a relatively small TCE of 27.8% is achieved for a 3.85 wt% of additive. XRD analysis reveals that the Fe_3O_4 particles are intercalated between the sheets of graphene, which may disrupt phonon transport occurring in the basal plane of graphene [144], explaining the relatively low TCE that the additive had. The composite has low enthalpies, which make it unsuitable for most PCM applications; it can be explained by the structure of the synthesised PCM which hinders the crystallisation of the molecules and is proved by XRD. The composite thermally degraded in two stages, one in the range 201.1–307.6 °C (due to polyurethane degradation) and the more significant in the range 332.7–434.5 °C (PEG degradation). The CPCM also showed minimal enthalpy losses after 100 thermal cycles.

Mishra et al. [81] surface functionalized GNPs with oleic acid and incorporated them into a hexadecane PCM. The authors of this work conducted an analysis of surface functionalization effects on thermal conductivity and have made a direct comparison between functionalized and unfunctionalized GNPs. Results from the unfunctionalized are discussed in Section 5.2 and can be seen in **Tables 6** and **8**. The results of the thermal conductivity of the f-GNPs compared with unfunctionalized GNPs show that for the same wt% the thermal conductivity of the f-GNPs is less, except at 0.01 wt% where it is larger. This can be explained by the fact that the surface coating may increase the thermal resistance of the interface by introducing phonon scattering centres. The larger thermal conductivity at 0.01 wt% could be attributed to better dispersion caused by improved forces/interaction between the nonpolar end of oleic acid and the nonpolar hexadecane; it is unclear whether the TCE increases beyond 0.01 wt% so further exploration should be required. It is probable that the stability and tendency to disperse well is held by the f-GNPs in higher values of wt%. Thermal cycling of both the f-GNP and GNP CPMCs showed that both experienced a reduction to 0% TCE in the liquid state TCE after only 4 cycles. The solid state TCE of the f-GNP CPCM initially decreased to 89.3% after the first cycle, then increased to 125% on the second cycle, followed by reductions on 3rd and 4th cycle to 110.7% on the 4th cycle. The unfunctionalized GNP experienced the same behaviour but with 104.3, 112.1, 80, and 77.9% TCE on the 1st, 2nd, 3rd· and 4th cycles, respectively. The stability of the f-GNP CPCM is better because of its reduced particle agglomeration.

Ge et al. [140] fabricated an epoxidized methoxy polyethylene glycol (EmPEG)/PDA-rGO (polydopamine-reduced graphene oxide). Like most other similar studies, as the wt% of filler increases the enthalpy of the composite decreases, as from analysis performed the spherulites of the PCM nucleate at the PDA-rGO inclusions. More numerous inclusions cause smaller crystals and more grain boundaries/imperfect crystals, thus resulting in a lower enthalpy. Also, the rule of mixtures will contribute to a lower enthalpy. After 100 thermal cycles the enthalpy and phase change temperature change are negligible.

Akhiani et al. [139] made rGO and GO from graphite flakes via a simplified hummers method, preceding its functionalization by adding and stirring oleylamine into a mixture of palmitic acid and

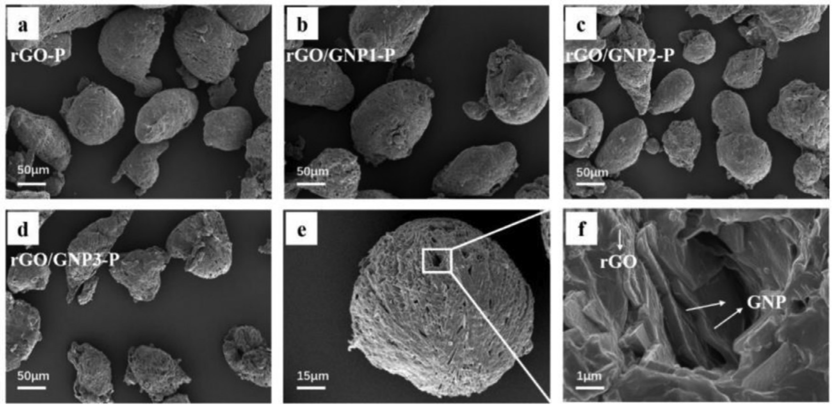

Fig. 15. SEM images of microencapsulated paraffin particles with different constitutions [145].

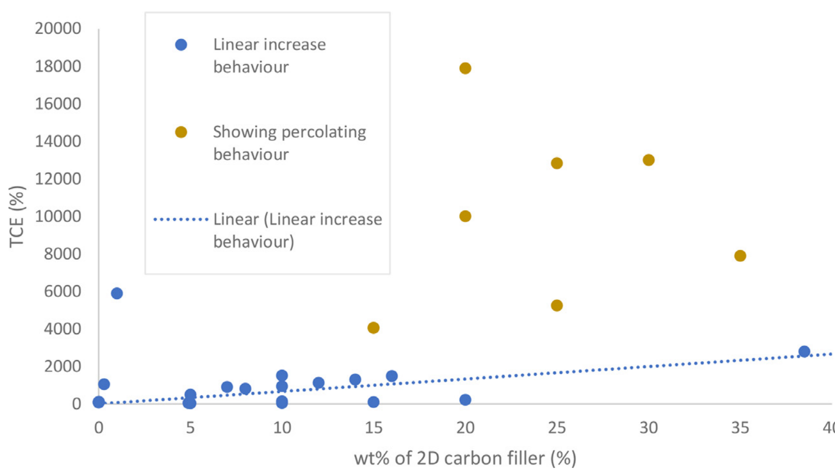

Fig. 16. TCE for different types of unfunctionalized 2D carbon fillers at varying wt% showing clear percolation behaviour.

rGO/GO followed by resting at 80 °C for 24 h. The composite experienced an increase of 8% in the phase change enthalpy value after 500 accelerated thermal cycles; the authors believe the cause of this is due to the increased protonation of the palmitic acid, allowing higher degrees of crystallization, with each cycle.

5.2.2. Encapsulation of 2D nanofillers

Encapsulation of the filling nanomaterials is a novel method to prevent various problems associated with PCM nano-composites, such as leakage and degradation [147,148], which the micro and nano capsules may be able to solve. Controlling the structure of the capsule is important so that the PCM does not leak out, stability is maintained and so that thermal conductivities are maximised. Table 9 displays some useful properties of various organic PCMs enhanced with encapsulated 2D carbon naomaterials and is ordered in decending TCE.

Dao et al. [146] made stearic acid latex particles using sodium dodecyl sulfonate (SDS) in varying amounts which were then coated with functionalized graphene to form a microencapsulated composite phase change material. The graphene self-assembles onto the latex particles due to the negatively polar SDS adsorbed onto the surface of the latex and the positive cationic surfactant hexadecyltrimethylammonium bromide (CTAB) attached to the graphene. SEM imaging shows a core-shell structure where the wrinkled sheets coat the latex particle. The authors also prepared a purely stearic acid -graphene composite and compared it against the stearic acid (latex)-CTAB graphene composite at the same loading of graphene; the prior melted to a liquid at 100 °C whilst the latter retained solidity and shape stability at the same temperature. Peak thermal degradation occurs at 292 °C and 282.3 °C for the microencapsulated composites containing SDS; compared to the temperature of 267.1 °C of pure stearic acid and 246.5 °C for stearic acid -graphene. This proves thermal stability of the microencapsulated form. Enthalpy values of the latex composite also outperformed the stearic acid -graphene for loading of graphene above 0.5 wt%, suggesting that the graphene additions do not disperse into the PCM matrix and exclusively coat the latex particles (which is confirmed by SEM). This is because interactions of the PCM at the surface of the filler have the possibility of reducing the degree crystallization of PCM [149].

Zhou et al. [150] characterized the effective thermal properties of a reduced graphite oxide-GNP/Paraffin CPCM finding it to be an effective method to enhance the thermal properties of the paraffin. It was found that the most effective ratio of graphite oxide to GNP was 7:3, resulting in an excellent TCE and a very high enthalpy retention. After 100 heating and cooling cycles the composite shows minimal heat flow losses on its DSC plot, proving its cycling stability. The thermal decomposition temperature of the paraffin can also be extended by increasing the wt% of GNP in the composite; the normal onset decomposition temperature of paraffin (140 °C) is extended by roughly 30 °C. **Fig. 15** illustrates the irregular and rough morphology of such encapsulated particles, which will become more irregular with higher ratios of GNP.

5.2.3. Summary of 2D nanofiller additions to PCMs

Fig. 16 shows clear evidence of thermal percolation when the filler wt% is larger than 20%, with the behaviour related to that predicted by the Lewis-Nielsen model [40] and the experimental

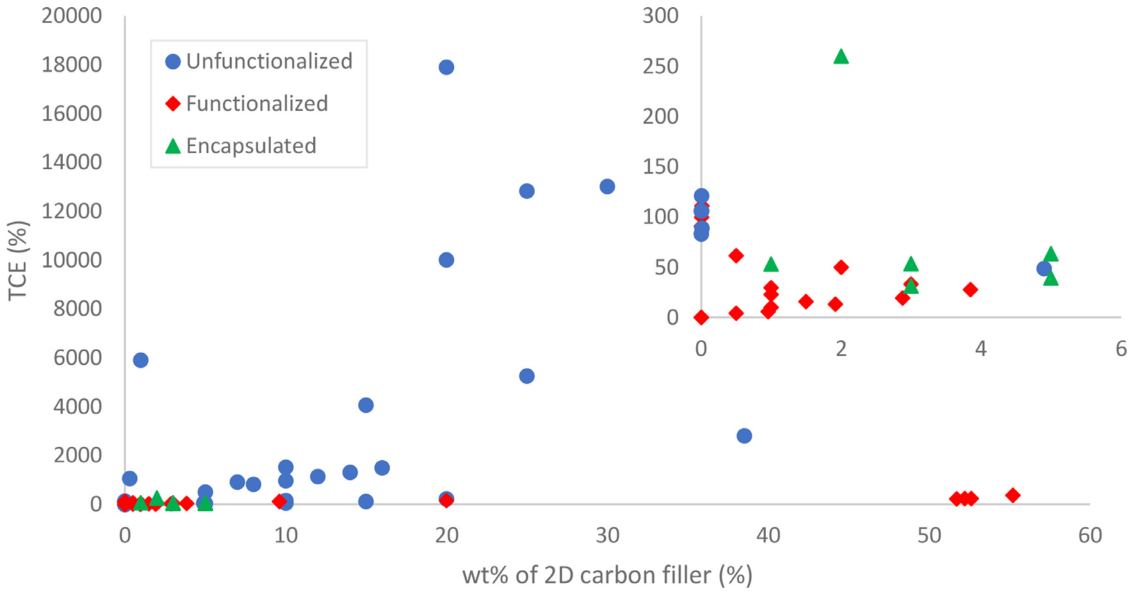

Fig. 17. TCE vs wt% of 2D filler for all types of nanofillers.

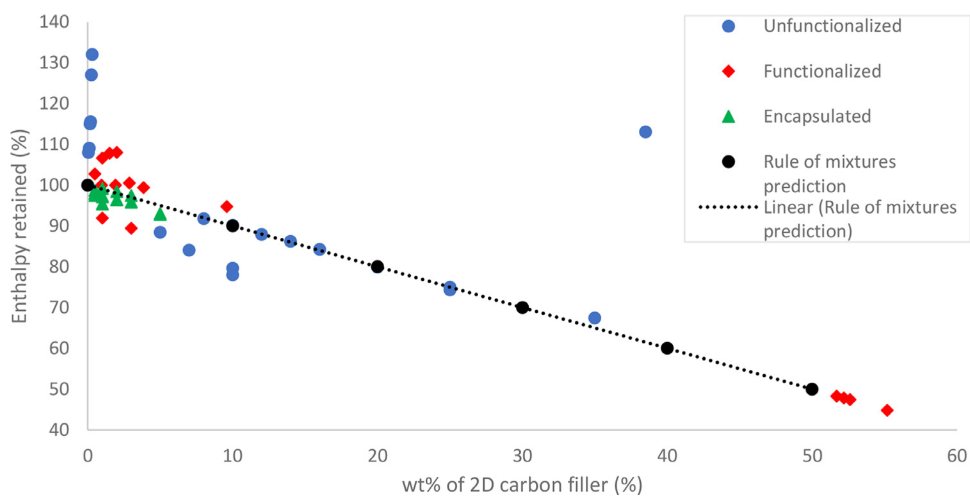

Fig. 18. Enthalpy retained vs wt% of different types of 2D carbon fillers.

data presented by Fariborz et al. [64]. Generally, there is a trend of increased TCE per wt% above 15 wt%, however there are also deviations and a wide distribution of values at higher values of wt%. This is likely because the data points are from different studies, using different fillers of different aspect ratio, resulting in thermal percolation occurring at different values of wt%. There are fewer studies published for higher values of wt% and due to this it is more difficult to predict the thermal conductivity of the composites. For low values of wt% the behaviour is more homogeneous, this is because the thermal conductivity is less dependent on the variations introduced by the different research conditions and is more dependent on the filler's inherent thermal conductivity and interfacial thermal resistance.

Fig. 17 evidently shows that a low wt% of functionalized 2D fillers have an increasing linear relationship with the TCE of the composite. There is less data available for high weight fractions of functionalized 2D carbon fillers, despite this it seems as if they do not perform well compared to the unfunctionalized fillers. It would be expected that functionalized fillers would outperform the unfunctionalized due to their better dispersion, allowing them to percolate at lower a wt%, however the available data does not support this. At higher wt% the functionalizing molecules may hinder phonon propagation between nanoparticles. Fig. 17 compares encapsulated 2D fillers to the other methods and shows that at a low wt% they compare closely in their TCE. Previous work could not be found to make comparisons with for the high wt% encapsulating 2D fillers.

The methods to achieve the best thermal properties of the CPCM via improvement of the unfunctionalized 2D carbon filling particle can be summarised as removing phonon scattering centres (making fillers more pristine), improving the interfacial coupling with the PCM, reducing the particle size to benefit from thermal conductivity enhancements via Brownian motion (however this is only for TCE in liquid state), and increasing the size and wt% of the filler.

Functionalization of the 2D filling particles imparts improved properties that are retained after cycling, leading to more a more practical CPCM. Enhanced dispersion, thermal degradation resistance, enthalpy retention and TCE retention after cycling; in addition to these special properties such as magnetic-to-thermal and solar-to-thermal energy conversions can be achieved by functionalization with certain methods. Functionalization may, however, in-

Table 10
Properties of 1D carbon nanofilled PCM composites.

1D Material	Sizes	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Ref]
GNFs	Length: 1 μm Diameter: 4–10 nm	Paraffin	5 wt%	11,820%	29.8	–	–	Liquid sonication	2008 [151]
SWCNT	–	Octadecane	0.25 wt%	292%	–	–	–	Liquid sonication	2013 [152]
a-CNT	Length: >10 μm	Paraffin	7 wt%	133.3%	0.56	218.56/-	111.2%	Solution impreg.	2019 [153]
a-CNT	Length: >10 μm Diameter: <10 nm	Paraffin	7 wt%	133%	0.56	218.56/-	111%	Impreg.	2019 [154]
CNT	–	Stearic acid	10 wt%	119.2%	0.57	176.2/173.7	83.5%/82.0%	Ball milling	2019 [155]
CNF	Length: 50 μm Diameter: 30 nm	Soy wax	10 wt%	44.8%	0.469	–	–	Melt mixing	2011 [156]
MWCNT	Length: 10–30 μm Diameter: 10–20 nm	PA and DMF	2.9 wt%	34.6%	0.203	–	–	Molten sonication	2019 [134]
MWCNT	Length: 1–10 μm Diameter: 5–20 nm	D-mannitol	0.5 wt%	32.8%	1.738	271.22/211.58	96.2%/96.4%	Liquid sonication	2018 [157]
CNT	Length: 1–10 μm Diameter: 5–20 nm	Paraffin	3 wt%	25.7%	0.567	137/141.4	90.3%/91.1%	Dispersant with sonication	2019 [158]
MWCNT	–	Capric acid	0.041 wt%	16.8%	0.08085	103.9/-	65.98%	Magnetic stirring	2020 [159]
CNF	–	Nonadecane-epoxy	2 wt%	10.2%	0.303	–	–	Melt mixing	2019 [160]

crease phonon scattering/interfacial resistance and it can be seen in **Fig. 17** that the TCE of the functionalized fillers mostly underperform the unfunctionalized fillers.

Encapsulation of PCMs using 2D carbon nanofillers achieves similar TCEs at a low wt%; and previous research exploring their use at high wt% could not be found. Encapsulation offers an enhancement on the thermal degradation temperature, increasing it, in accompaniment to cycling stability.

In **Fig. 18** it can be seen that the unfunctionalized fillers mostly follow the rule of mixtures prediction, except in the low wt% cases below 5 wt% where the fillers aided in the PCM crystallization; this data comes from Putra et al. [133]. Encapsulation using 2D filler retains less enthalpy than predicted, which is expected due to the fact that other materials are added, such as SDS, in order to create latex particles; this will subtract from the latent heat of solidification values. The limited data from functionalized fillers suggests they have the possibility to retain composite enthalpy well, likely due to enhanced crystallinity caused by alike polarity interactions.

5.3. 1D carbon nanofillers

Table 10 displays the properties of various organic PCMs loaded with different 1D carbon nanomaterials and is ordered in decending TCE. Harish et al. [152] found that an enhancement of 250% could be made on the solid state thermal conductivity of octadecane with 0.25 wt% SWCNTs and a very anomalous 10% TCE in the liquid state. Thermal cycling showed that the behaviour is reversible. The authors propose theories on why this occurs with the most plausible being that the CNTs are pushed to the grain boundaries when the PCM crystallizes, forming a percolating network of CNTs along grain boundaries, similar to that mentioned by Mishra et al. [81]. Xuehong et al. [158] prepared a CNT/Paraffin CPCM and did not find the same high solid thermal conductivity and low liquid thermal conductivity behaviour, this may be because of the dispersant 'WinSperse3050' added by Xuehong et al. which acts as a surfactant on the CNT and prevents clumping of CNTs, aiding to the liquid state thermal conductivity. Congruent with most other previous works, the thermal conductivity increases with increased wt% and the heat capacity decreases due to the rule of mixture [158].

Salyan et al. [157] used MWCNTs to enhance the thermal conductivity of d-mannitol; a linear increase in thermal conductivity is achieved for linear increases in wt%. The maximum TCE of 132% at 0.5 wt% was achieved by sonicating the mixture above the melting point of the d-mannitol. After 100 melt cycles the enthalpy of the 0.5 wt% had reduced by 19.12%; images show that the d-mannitol has lightly browned, suggesting its oxidation and the reason for the reduction. Hence, the choice of the PCM should account for oxidation as well.

Weinstein et al. [151] explored the addition of different morphology graphite nano-fibres (GNFs) to a paraffin wax at different loadings. Herringbone, platelet, and ribbon GNFs at a loading of 5 wt%, shown in **Fig. 19**, were used, with resulting thermal conductivities of 17.2, 25.3 and 29.8 W/mK, respectively. The ribbon GNFs had the highest TCE due to phonon propagation being enhanced in the axial direction caused by the alignment of the graphite, which has a higher thermal conductivity in its 2D plane direction. Whilst the ribbon had the highest TCE, the entirety of the PCM had not melted, wasting the heat storage capabilities. Although the increased loading of GNF improved the thermal conductivity of the CPCM, it also reduced the convective heat transport within the composite and aided in overheating at the bottom layer. The authors explored the optimal loading for thermal spreading and reduction in thermal gradients for enhanced temporal PCM effectiveness; and they found that herringbone at 3.75 wt% improved the heat penetration and delayed the onset of the steady state. It is important to note that heat capacities of the CPCM were minimally affected when compared to the base PCM.

Yadav et al. [159] prepared a MWCNT/capric acid CPMC via a magnetically stirring hot plate. A small enhancement was made with the CPCM having 131.29% of the original PCM thermal conductivity for only a small, 0.02vol%, fraction of filler. The melting enthalpy of the composite for 0.02vol% of MWCNT was 103.9, and only 65.98% the original value of the pure PCM. Following the rule of mixtures and the trends in other studies, the enthalpies will de-

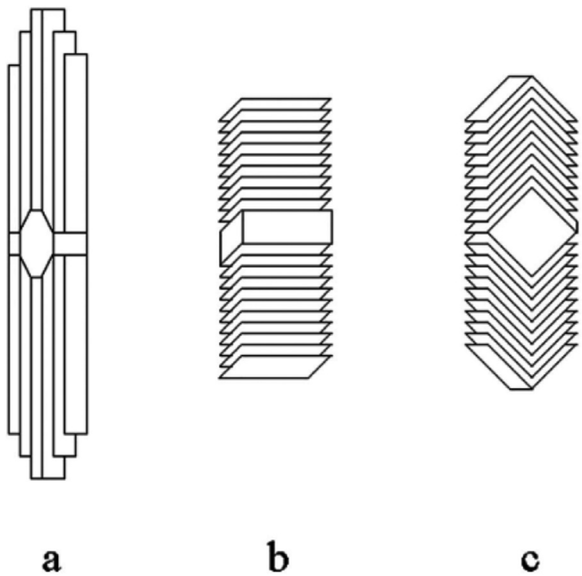

Fig. 19. GNF morphologies, (a) ribbon, (b) platelet and (c) herringbone[151].

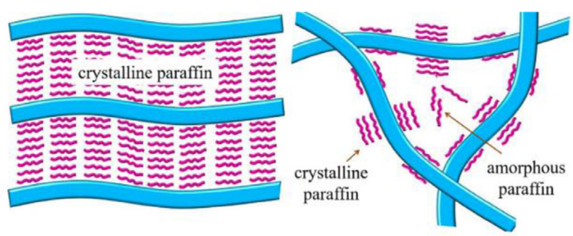

Fig. 20. Illustration of enhanced paraffin crystallisation around aligned CNTs [154].

crease linearly for increasing additions of filler; this means that the resulting CPCMs enthalpy will not be suitable for most applications for a higher vol% of MWCNTs, which is necessary to achieve a practical thermal conductivity. Unlike other studies, the authors of this study looked at the increase in convective thermal transport and found that the largest increase, 13.8%, was obtained for 0.02vol% MWCNTs.

Zhu et al. [154] explored the effect of aligned CNTs on CPCM thermal properties and heat capacity when compared to randomly dispersed CNTs. The results show that the aligned CNT CPCM has a 19.6% increase in melting enthalpy when compared to the base PCM (see Fig. 21). Due to CNTs having π electrons on their surfaces, it is proposed that C–H—-π interactions are made between the paraffin C–H bonds and the CNT π electrons, which enhance the paraffin alignment, promoting higher degrees of crystallisation and thus enhancing the melting enthalpy. As illustrated in Fig. 20, the aligned CNTs promote higher degrees of PCM crystallinity, this is due to the randomly dispersed CNTs higher likelihood of CNT crossover, preventing PCM crystallisation at the crossover points. These theories are introduced and characterized by Zhu et al. with polarizing microscopy and cryo-electron microscopy. The authors also explored the thermal stability, finding that after 500 cycles the morphology of the aligned CNTs were still intact and the paraffin had not phase separated; leakage of paraffin was not observed until a paraffin was at 95 wt% which was most likely caused by capillary forces introduced by aligned CNTs.

5.3.1. Functionalisation of 1D nanofillers

Table 11 displays the properties of various PCMs loaded with different functionalized 1D carbon nanomaterials and is ordered in decending TCE. Cao et al. [161] looked at the viability of hexadecyl acrylate (HDA functionalized MWCNT and SWCNT for a direct comparison of their thermal properties, which is useful to understand the effect of aspect ratio and grafting rates. Thermal conductivity enahncements of 339% and 134% were achieved for the f-MWCNTs and f-SWCNTs, respectively, for the same wt% fraction of each in a HDA matrix. Per weight percent of f-SWCNT, there will be more covalent bonds between the HDA and SWCNT when comapared to the f-MWCNTs, due to the nature of the MWCNT and its mulitple walls being inaccessable to the HDA molecule and thus to react with. Due to the larger fraction of grafting per wt% on the SWCNT, more phonon scattering centres have been created, lowering the thermal conductivity when compared to the f-MWCNT composite. Specific heat values of 1.089 and 0.629 MJ/(m^3K) were recorded for f-SWCNT and f-MWCNT composites respectively, with the MWCNT CPCM exhibiting a lower value due to the larger it occupies per wt%, relating to the rule of mixtures and how the heat capacity of the CNT is low. Also, after 100 thermal cycles, both the f-SWCNTs and f-MWCNT composites exhibited negligible losses in enthalpy values. Thermal stability was displayed for both types of f-CNT with a 5% mass loss occuring at temperatures over 300 °C; f-MWCNT shows a higher thermal stability temperature at 5% mass loss, which is surprising due to its lower grafting rate and thus, van der waals interactions between the grafts and the matrix.

Ranjbar et al. [84] explored the effect of different functionalizations on MWCNTs in different polarity PCMs; paraffin (nonpolar), stearic acid (slightly polar) and polyethylene glycol (polar) were used. The pristine MWCNTs are nonpolar and the COOH and OH functionalzied MWCNTs are polar. The results from testing the polar and non-polar MWCNTs with the polar and nonpolar PCMs shows that nonpolar-nonpolar composites have less sedimentation and increased physical stability. MWCNT polar functionalizations do not significantly contribute to the stability and will sediment after few cycles. Stearic acid behaved much like paraffin because of its dominant nonpolar molecule ‘head’, which has a much larger molecular mass. Nonpolar f-MWCNTS (functionalized-MWCNTs) performed worse in stearic acid because of its larger polarity when compared to paraffin. As expected, using polyethylene glycol as the polar PCM produced better stability results in combination with the polar f-MWCNTs and also with additions of surfactant (SDBS). However, the authors did not share the results of the other f-MWCNT/PCM composites and only gave the results from MWCNT-OH functionalized/PEG composite which was reported to have the best thermal conducitivity enhancement in the liquid stage of 116.58% and in the solid state was roughly 112.4%.

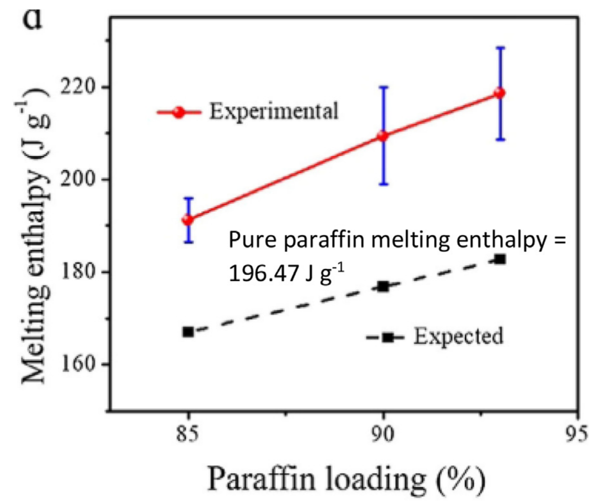

Fig. 21. Graph of melting enthalpy vs loading for a paraffin/a-CNT composite [154].

Table 11
Properties of differently functionalised 1D nanofilled PCM composites.

1D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Reference]
f-MWCNT	Length: 0.5–2 μm Diameter: <8 nm	Hexadecyl acrylate	1 wt%	339%	0.877	–	–	Sonication and reaction	2019 [161]
f-SWCNT	Length: 0.5–2 μm Diameter: 1–2 nm	Hexadecyl acrylate	1 wt%	134%	0.4675	–	–	Sonication and reaction	2019 [161]
f-MWCNT	Length: 10–30 μm Diameter: 10–30 nm	Beeswax	5 wt%	84%	0.46	115.54/126.20	75.4/ 81.9%	Solvent sonication	2018 [162]
f-MWCNT	Length: 3–12 μm Diameter: 8–15 nm	Polyethylene glycol	5 wt%	80%	0.531	150/-	90.9%	Solvent sonication	2019 [163]
OI-CNT	Length: 10–30 μm Diameter: 5–15nm	C18/C28	0.8 wt%	44.3%	0.329	187.9/-	70.2%/-	Sonication and reaction	2020 [164]
f-MWCNT	Length: 10–30 μm Diameter: 10–20 nm	Polyethylene glycol	1 wt%	16.82%	Approx. 0.32	–	–	Solvent and melt sonicated	2020 [84]

Table 12
Properties of differently encapsulated 1D carbon nanofilled PCM composites.

1D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Reference]
OI-CNT	CNT Length: 10–30 μm CNT Diameter: 5–15 nm	C18/C28	0.8 wt%	44.3%	0.329	187.9/-	70.2%/-	Sonication and reaction	2020 [164]

Wang et al. [163] prepared a f-MWCNTs (functionalised with sodium dodecyl sulfate (SDS)) - PEG8000 CPCM via sonicating the mixture with water as a solvent, followed by vacuum evapouration of the water. Weight fractions of 0.5, 1, 2, and 5% were used to make the samples. From 1 to 5% there was no surface leakage, even at temperatures far over the melting point of the PCM, showing that SDS f-MWCNTs have good form-stable properties. The surfactant increases the intermolecular forces between PCM and f-MWCNT, in addition to capillary forces created by the porous MWCNT morphology. After running one hundred accelerated heating and cooling cycles ranging from 0 to 100 °C, the composite only had a 0.9% reduction in its latent heat of fusion, which is negligible and is a promising value for use in applications.

Rawi et al. [162] functionalized MWCNTs via reaction with nitric and sulfuric acids, which were then sonicated and stirred in an ethanol solvent before evapouration of the solvent. FTIR results show that C=O, -COOH and C–O bonds exist within the f-MWCNT provided that oxidation has occurred. Similar results were obtained by Wang et al. [163] with an almost identical TCE for the same wt% of similar filler. The 4% larger TCE obtained by Rawi et al. could be due to the aspect ratio of their f-MWCNTs being larger, creating more interactions between f-MWCNTs and percolating thermal transport. Rawi et al. did have a larger reduction in enthalpy values when compared to Wang et al., possibly because of the chemistry of the beeswax molecules which consists of many compounds and different esters. This might cause a disruption in the crystallinty of the wax and thus enthalpy of melting as the polar nature of the MWCNTs functional groups would attract the polar ester group in the centre of the molecule.

5.3.2. Encapsulation of 1D nanofillers

Table 12 displays the properties of a PCM loaded with an encapsulated 1D carbon nanomaterial. Combining carbon nanomaterial functionalization effects with encapsulation effects can be done to improve the thermal conductivity of encapsulated PCMs. Meng et al. [164] grafted CNTs with octadecyl isocyanate (OI) in an attempt to improve their compatibility and dispersion within the matrix PCM, n-octadecane/n-octacosane (C18/C28), followed by encapsulating them in a melamine-formaldehyde shell. The microencapsulation methods by themselves lowered the thermal conductivity by 15.8% (without OI-CNT); with OI-CNT the TCE was increased by 44.3% compared to pure polymer and 71.4% compared to the encapsulated pure polymer.

5.3.3. Summary of using 1D nanofillers with PCMs

Fig. 22 shows evidence of linear enhancements in TCE for both functionalized and unfunctionalized fillers at low loadings; as the data comes from different research papers using fillers of different morphology there is a high amount of scatter on the graphs. It appears as if the functionalized filler outperform the unfunctionalized, however due to the variation mentioned before and the low amount of data at high loadings, conclusive statements cannot be made. Shown by the graph inset is the same data, but including the results by Weinstein et al. [151], whose TCE for using GNFs of varying morphology greatly outperforms the other 1D fillers; this is possibly due to the enhanced phonon propagation allowed by the GNF morphology. There is a lack of research for both functionalized and unfunctionalized 1D carbon fillers at high filling fractions (wt%> 10) therefore understanding how the functionalization of 1D fillers affects the percolation threshold is not possible; although it is likely that functionalization will lower the percolation threhold due to increased particle dispersion within the PCM.

Fig. 23 shows the enthalpy values of the composites with functionalized and unfunctionalized 1D fillers. The composites mostly underperform when compared to the predictions by the rule of mixtures; this could be due to fillers reducing the extent of crystallization of the PCM and thus the enthalpy. In the cases where the composites retain more enthalpy than the rule of mixture predicts, the fillers provide a site for the alignment of PCM molecules, aiding the formation of PCM crystals with higher enthalpy values. Functionalized filler seem to retain more enthalpy than unfunctionalized fillers that do not aid PCM crystallization, this could

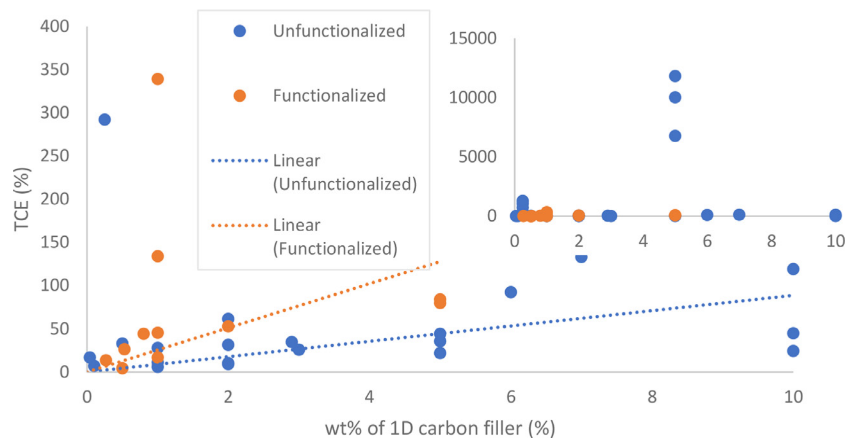

Fig. 22. TCE vs wt% for functionalized and unfunctionalized 1D fillers, with main graph disregarding results from Weinstein et al. [151] and graph inset including their results.

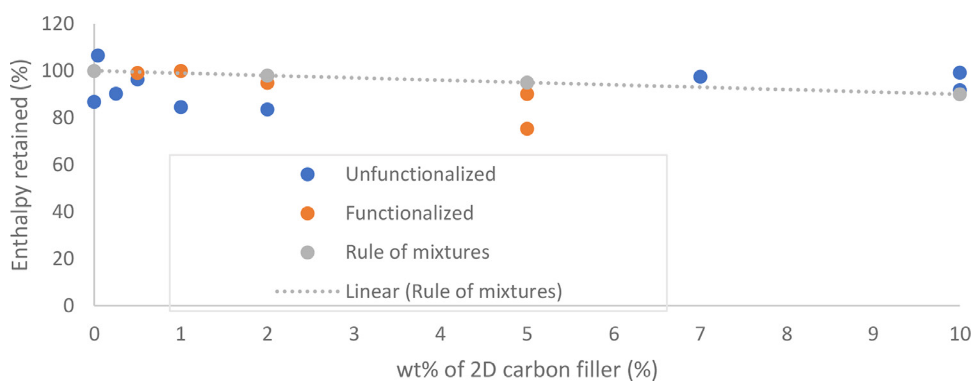

Fig. 23. graph of enthalpy retained vs wt% of 1D filler.

Table 13
Properties of 0D carbon nanofilled PCM composites.

0D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Reference]
CBNP	Diameter: 21 ± 2 nm	Hexadecane	0.01 wt%	121.4%	0.551	–	–	Sonication and stirring	2018 [81]
CBNP	Diameter: 30 nm	Palmitic acid and DMF	0.6 wt%	67.2%	0.252	–	–	Liquid phase sonication	2019 [134]

be due to functionalized fillers experiencing less ‘crossover’ due to their dispersion (see Fig. 20) [154].

5.4. 0D carbon nanofillers

Table 13 displays the properties of different organic PCMs loaded with 0D carbon nanomaterials and is ordered in decending TCE. Mishra et al. [134] made a composite using a PCM blend of palmitic acid and di-methyl formamide (DMF). With larger wt% of DMF the onset phase change temperature would lower, however as DMF inclusions increased the latent heat of the blend would significantly lower, thus showing limited tunability of the phase change temperature. The thermal conductivity of the PCM-CBNP composite varied significantly at different states; for liquid, phase transition and solid states the TCE were 11.1, 146.9 and 67.2% respectively for 0.6 wt% of CBNP. The differences in TCE per state must be understood to properly design a composite PCM; it is likely that crystallisation events in the PCM will alter the dispersion of the filler nano-inclusions, changing percolation behaviour. Composite thermal conductivity reduces after every thermal cycle. The CBNP inclusions would aggregate after multiple thermal cycles, creating many interfaces with acoustic mismatches, leading to interfacial thermal resistance and phonon scattering; also, the percolation of CBNP would be less probable and thus explains the poor cycling stability.

Mishra et al. [81] achieved a relatively high TCE in the solid state for a low wt% of carbon black nano powder. In the liquid state the TCE was 6.4%, in the solid state was 121.4% and, in the liquid-solid transition state was 300%, all fractions at 0.01 wt%. These variations can be explained by the morphology created by the PCM, hexadecane, which forms crystalline regions when solidifying that will push nano inclusions to its grain boundaries. The reason for why the transition region TCE is much higher than the solid TCE is due to residual stress introduced by the PCMs solidification, which act to break up the crystalline structure [165] and hinders phonon propagation. Nano inclusions at the grain boundaries will form percolating networks at lower wt% than if fully dispersed. Although this 0D particle is less capable of efficiently transferring heat through percolating networks due to its aspect ratio, the mechanisms of solidifying in the PCM can create beneficial nano morphologies, enhancing the thermal conductivity. In the same study, GNPs of higher thermal conductivity (~3000 W/mK

Table 14
Properties of functionalised 0D carbon nanofilled PCM composites.

0D Material	Size	Matrix PCM	Loading of filler	TCE (%)	Solid thermal conductivity (W m^{-1} K^{-1})	Melting/Freezing Enthalpy (kJ/kg)	Enthalpy retained (%)	Preparation method	Year [Reference]
MCMS	Pore size: 3.4 nm	PEG	30 wt%	21.1%	0.304	112.08/107.34	58.9/61.3%	Vacuum impregnation	2019 [167]

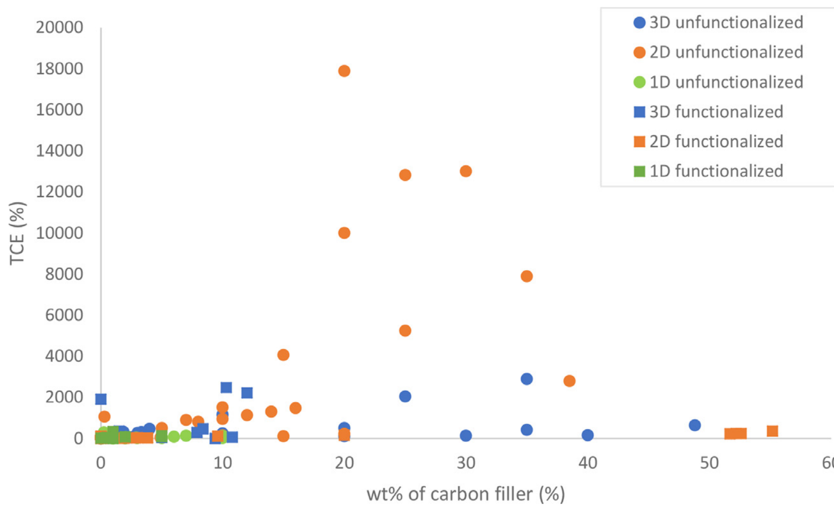

Fig. 24. Comparison of TCE vs wt% for all dimensions of filler.

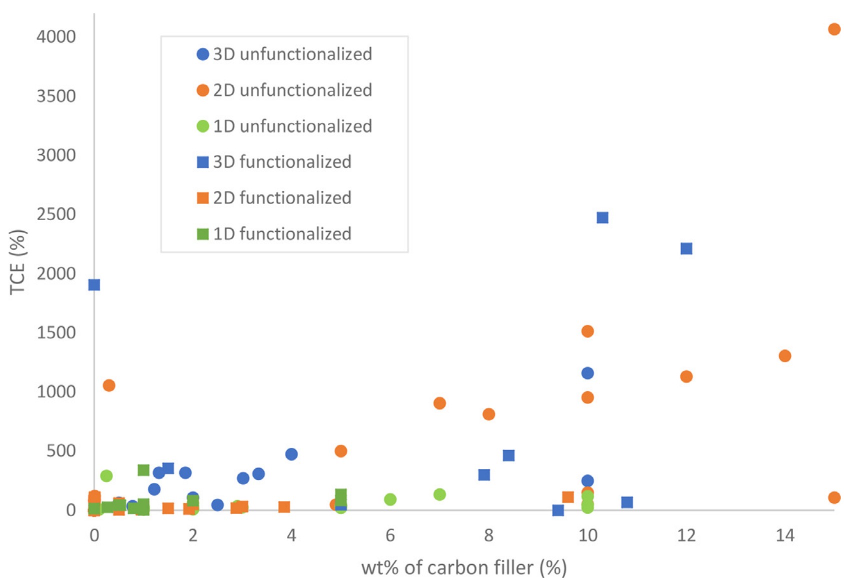

Fig. 25. Comparison of TCE vs wt% for all dimensions of filler at medium filling fraction.

compared to ~0.4 W/mK of CBNP) created a lower TCE in the composite and this is due to the fact that CBNP particles form close packed aggregates which will have improved interaction with other aggregates, increasing phonon propagation [166].

5.4.1. Functionalization of 0D carbon nanofillers

Table 14 displays the properties of an organic PCM loaded with a functionalized 0D carbon nanomaterial. Chen et al. [167] fabricated mesoporous carbon microspheres (MCMS) via pyrolysis of polydopamine microspheres. The MCMS had C–OH and C=O functional groups, confirmed by XPS. The functional groups allowed for hydrogen bonding between the PEG and MCMS and SEM imaging showed a gelatinized surface with colloidal partial structure, which suggests that the PCM is fully absorbed on the MCMS. However, aggregation is obvious and has probably reduced the potential TCE of the composite; improved dispersion would likely result in a higher TCE. The average pore sizes of the MCMS were 3.4 nm which are large enough for the PEG to diffuse inside the spheres; PEG embedding was confirmed via XRD, also the crystallization fraction was shown to not be influenced. TGA analysis showed that pure PEG would begin to degrade at a temperature of 200 °C and be fully degraded at 420 °C; the CPCM would begin to degrade at 350 °C which confirms the thermal stability and thermal resistance of the composite. Due to the low TCE and low enthalpy retention for the relatively high weight fraction used this filler is not suitable for use as more suitable results can be achieved for other fillers.

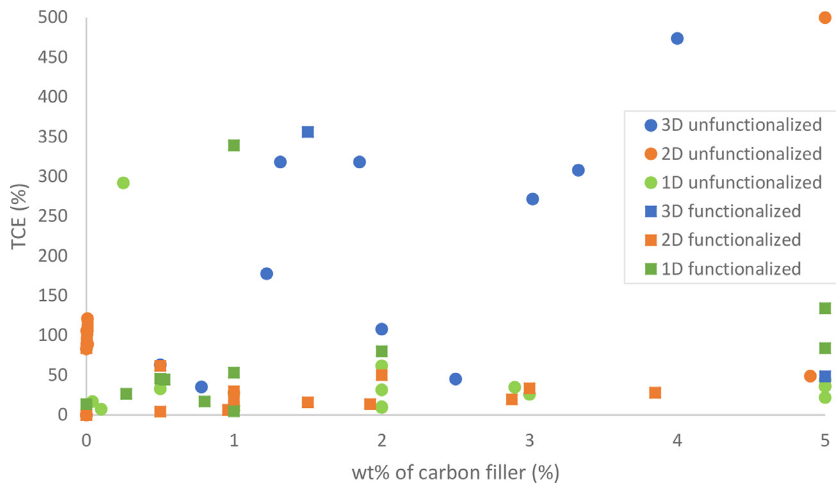

Fig. 26. Graph of TCE vs wt% for all types and dimensions of carbon fillers at a low filling fraction.

5.5. Comparison of different carbon nanofillers

Figs. 24, 25, and 26 are ignoring the data taken from Weinstein et al. [151] as it dwarfs the other sets of data, concealing the trends of the data sets. Fig. 24 demonstrates the large TCE that can be achieved via creating percolating networks of high wt% 2D fillers.

Fig. 25 demonstrates the need for more data; as it is hard to determine whether functionalized 2D and 3D fillers perform better than unfunctionalized 2D and 3D fillers, respectively.

In Fig. 26 it can be seen that a low wt% of carbon filler 3D structures impart the highest TCE on the composite at roughly 300%, whereas all other dimensions of carbon filler are generally below 100%. The performance of 3D carbon nanofillers at a low wt% is three orders of magnitude higher; this is probably due to the morphology of the 3D filler, which will form interconnected networks at lower weight fractions, allowing phonon propagation. At low weight fraction the phonon transport in 2D and 1D fillers is hindered by the interfaces between filler and PCM which will scatter phonons.

Figs. 13, 18 and 23 are the graphs of enthalpy retained for 3D, 2D and 1D fillers respectively; they can be compared for their values. It can be seen that 2D fillers are able to give the largest value of enthalpy retained, which occurs at a low wt% of unfunctionalized filler. The enthalpy retained is able to be improved beyond the enthalpy of the base material by improved PCM crystallization. 3D and 1D fillers closely represent the rule of mixture for their enthalpy retained but have the possibility to improve crystallization properties. However, a conclusion cannot be drawn from the data about whether functionalization of the 3D or 1D filler improves the enthalpy retained.

6. Recommendations for future works

It would be beneficial for researchers wishing to design and understand the most effective filling materials to use an established standardised method to rank the enhancements of different fillers and composites against each other. A cost benefit analysis should also be done to assess which filling particle and PCM will give the best cost-to-enhancement of the desired property, and if the gains made from the utilisation of the CPCM are higher than the costs of implementation; and this can be recommend for future works. This would require authors to follow a protocol and to carefully characterise the aspects of a composite and its filler according to fundamentally useful qualities and properties such as:

-Thermal conductivity enhancement

-Enthalpy retained

-Weight/fraction

-Filler dimension properties (length, width, and surface area of filler particles)

-Solid and liquid composite thermal conductivity

-Properties at X amount of thermal cycles (with X to be a unanimously agreed upon value)

-Cost of materials and assembly

These properties should be displayed in a table that is easy to find; this would help future researchers to understand and utilise the physical phenomena effectively, whilst producing results that have clear, unambiguous causes. For example, some authors will not characterise the dimensions or image the particles in the PCM matrix, making it difficult for others to understand whether the property enhancements are due to the percolation or other phenomena. Making all the data available would allow other researchers to draw conclusions that the original researchers may have missed. Some authors will often omit fundamental characteristic properties of the CPCM or filler, or not present their results concisely. Most authors do a good job of this; however, it can still be a laborious task to compare the effectiveness of different fillers. If experimental numerical data was supplied in an article's appendix, in an easily utilisable manner, research and review of the topic would become faster and more efficient.

Often a high solid state TCE will be reported, and then authors will omit the liquid state thermal conductivity, creating an incomplete understanding of a composite's performance. Thermal conductivity enhancements reported by authors may not consider the effect of convective transport and the effect higher wt% of filler has on CPCM viscosity.

To understand the thermal percolation in 1D carbon nanofiller composites, research at weight fractions above 10% is needed and more study of 2D fillers is also needed at a high wt%. More work should also be carried out in researching the effect of 0D carbon nano particles in CPCMs, however it is not clear the benefit that they would impart because of their small aspect ratio.

Encapsulation is an approach used to improve the leakage and stability of CPCMs and will impart similar properties to that of 3D filling nanomaterials alone. Hence, more research should be conducted to explore the effect of encapsulating different dimension nanomaterials and compare them to non-encapsulating methods to assess the benefits and drawbacks of each method.

Research should be conducted that explores the TCE at a wide range of weight fractions (which spans before and after the predicted thermal percolation threshold). It should investigate a single type of filling nanomaterial, so that factors contributing to the TCE, such as geometry, can be ignored. This would be useful to confirm the thermal percolation threshold in CPCMs. In addition, characterisation should be performed to understand the microstructural changes and chemical behaviour of the composites at key points of interest, such as the thermal percolation threshold, after thermal cycling, and to assess chemical interactions with the CPCM container.

7. Conclusions

In order to make renewable energy solutions such a latent heat thermal energy storage widespread, the relevant research must be made easily available and consumable for researchers. Outlined in this project is the physical phenomena that influence the composite phase change materials properties such as: thermal conductivity enhancement, enthalpy retention, and thermal stability. The important phenomena are interconnected, influencing each other, and can be summarised as filler geometry, wt% of filler, interfacial chemistry, thermal percolation, and phonon propagation. The thermal conductivity of available organic PCMs must be improved upon by the addition of highly conductive additives; here, carbon nanomaterials of 1D, 2D and 3D have been proven to do so. Unsurprisingly, there are large variations in the sets of data, even within one material dimension group. However, trends can be seen when comparing dimensional groups. At low values of wt% ($< 5\%$) 3D carbon nanomaterials impart the highest thermal conductivity enhancement (roughly 300%) when compared to that of lower dimensional materials (roughly 50%). 3D fillers have a positive linear relationship between wt% and composite TCE. 2D fillers impart the highest TCE on the composite at high values of wt% and they typically create thermally percolating networks above 15 wt%. Researchers would benefit if there was more data available for the encapsulation of PCMs using 1D carbon nanofillers and for 1D fillers at high values of wt%.

The composite latent heat values are dependent on the fraction of PCM in the composite and the fraction of crystallized PCM also; their values can be improved upon, going beyond the original pure PCM value, which was previously not thought possible. From data collected in this review it can be seen that 2D fillers improve the composite latent heat the most. If the filler acts as a site for the alignment of PCM molecules, then crystallization degree will be increased and hence, latent heat. Filler functionalization can lower the aggregation degree by improving the intermolecular interactions with the PCM, making particle-particle interactions less favourable; it can also help to prevent leakage, maintain TCE and thermal stability after cycling. Furthermore, functionalization can also add novel properties to the composite such as magnetic and solar-to-thermal energy conversion, and there exist other methods to improve properties such as composite compression into blocks. Overall, the thermal properties of various carbon additives in a wide variety of PCMs have been reviewed and compared, with their causal properties identified.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] M. Abdul, B. Saad, S. Muhammad, Limitations, challenges, and solution approaches in grid-connected renewable energy systems, Int. J. Energy Res. 44 (Issue 6) (2020) 4132–4162, doi:10.1002/er.5033.

[2] Sørensen B., Renewable Energy (4th Edition), 2011. 3–34. 10.1016/B978-0-12-375025-9.00001-4.

[3] K. Lovchinov, H. Nitchev, M. Petrov, R. Stoykov, N. Tyutyundzhiev, Simulation and modelling of small solar thermal plant, Measurement 162 (2020) 107944 1 October, doi:10.1016/j.measurement.2020.107944.

[4] Ogbonnaya C., Turan A., Abeykoon C. Energy and exergy efficiencies enhancement analysis of integrated photovoltaic-based energy systems. J. Energy Storage, 26, December 2019, 101029. 10.1016/j.est.2019.101029.

[5] T. Berger, R.L. Czudaj, Commodity futures and a wavelet-based risk assessment, Phys. A: Stat. Mech. Appl. 554 (2020) 124339 15 September, doi:10.1016/j.physa.2020.124339.

[6] C.H. Dasanayaka, C. Abeykoon, P. Nagirikandalage, Modelling of the effects of renewable energy establishments towards the economic growth of a nation, in: Proceedings of the 7th International Conference on Fluid Flow, Heat and Mass Transfer. Proceedings of the 8th International Conference on Fluid Flow, Heat and Mass Transfer, 2021 21/05/21.

[7] K. Du, J. Calautit, Z. Wang, Y. Wu, H. Liu, A review of the applications of phase change materials in cooling, heating and power generation in different temperature ranges, Appl Energy 220 (2018) 242–273 15 June, doi:10.1016/j.apenergy.2018.03.005.

[8] M. Fatih Demirbas, Thermal energy storage and phase change materials: an overview, Energy Sources, Part B Econ. Plan Policy 1 (2006) 85–95, doi:10.1080/009083190881481.

[9] I. Sarbu, C. Sebarchievici, A comprehensive review of thermal energy storage, Sustainability 10 (2018) 191, doi:10.3390/su10010191.

[10] F. Chen, R. Huang, C. Wang, X. Yu, H. Liu, Q. Wu, et al., Air and PCM cooling for battery thermal management considering battery cycle life, Appl. Therm. Eng. 173 (2020) 115154 5 June, doi:10.1016/j.applthermaleng.2020.115154.

[11] D. Zou, X. Ma, X. Liu, P. Zheng, Y. Hu, Thermal performance enhancement of composite phase change materials (PCM) using graphene and carbon nanotubes as additives for the potential application in lithium-ion power battery, Int. J. Heat Mass Transf. 120 (2018) 33–41, doi:10.1016/j.ijheatmasstransfer.2017.12.024.

[12] L. Fan, J.M. Khodadadi, Thermal conductivity enhancement of phase change materials for thermal energy storage: a review, Renew. Sustain. Energy Rev. 15 (2011) 24–46, doi:10.1016/j.rser.2010.08.007.

[13] H. Badenhorst, A review of the application of carbon materials in solar thermal energy storage, Sol. Energy 192 (2019) 35–68, doi:10.1016/j.solener.2018.01.062.

[14] Yang G., Yim Y., Lee J.W., Heo Y., Park S. Carbon-filled organic phase-change materials for thermal energy storage : a review 2019. Molecules 2019, 24(11), 2055; 10.3390/molecules24112055.

[15] B. Eanest Jebasingh, A. Valan Arasu, A comprehensive review on latent heat and thermal conductivity of nanoparticle dispersed phase change material for low-temperature applications, Energy Storage Mater. 24 (2020) 52–74, doi:10.1016/j.ensm.2019.07.031.

[16] E. Gariboldi, L. Colombo, D. Fagiani, Z. Li, Methods to characterize effective thermal conductivity, diffusivity and thermal response in different classes of composite phase change materials, Materials 12 (16) (2019) 2552, doi:10.3390/ma12162552.

[17] Y. Lin, Y. Jia, G. Alva, G. Fang, Review on thermal conductivity enhancement, thermal properties and applications of phase change materials in thermal energy storage, Renew. Sustain. Energy Rev. 82 (2018) 2730–2742, doi:10.1016/j.rser.2017.10.002.

[18] S. Wu, T. Yan, Z. Kuai, W. Pan, Thermal conductivity enhancement on phase change materials for thermal energy storage : a review, Energy Storage Mater. 25 (2020) 251–295, doi:10.1016/j.ensm.2019.10.010.

[19] K. Pietrak, T. Wiśniewski, A review of models for effective thermal conductivity of composite materials, J. Power Technol. 95 (2015) 14–24.

[20] H. Nazir, M. Batool, F.J. Bolivar Osorio, M. Isaza-Ruiz, X. Xu, K. Vignarooban, et al., Recent developments in phase change materials for energy storage applications: a review, Int. J. Heat Mass Transf. 129 (2019) 491–523, doi:10.1016/j.ijheatmasstransfer.2018.09.126.

[21] B.E. Jebasingh, A.V. Arasu, A detailed review on heat transfer rate, supercooling, thermal stability and reliability of nanoparticle dispersed organic phase change material for low-temperature applications high density polyethylene, Mater Today Energy 16 (2020) 100408, doi:10.1016/j.mtener.2020.100408.

[22] J.W. Ramsey, E.D. Larson, E.M. Sparrow, Freezing on a finned tube for either conduction-controlled or natural-convection-controlled heat transferInternational, J. Heat Mass Transf. 24 (2) (1981) 273–284 February, doi:10.1016/0017-9310(81)90035-1.

[23] L. Fan, H. Lu, Heat transfer during constrained melting of nano- enhanced phase change materials in a spherical capsule : an experimental study 2016, J. Heat Transfer. 138 (12) (2016) 122402 Dec, doi:10.1115/1.4034163.

[24] H. Chen, S. Li, P. Wei, Y. Gong, P. Nie, X. Chen, et al., Cells Experimental study on characteristics of a nano-enhanced phase change material slurry for low temperature solar energy collection, Sol. Energy Mater Sol. Cells 212 (2020) 110513, doi:10.1016/j.solmat.2020.110513.

[25] S.C. Kim, R. Prabakaran, D. Sakthivadivel, N. Thangapandian, A. Bhatia, P.G. Kumar, Thermal transport properties of carbon-assisted phase change nanocomposite, Fuller. Nanotub. Carbon Nanostruct. 28 (2020) 925–933, doi:10.1080/1536383X.2020.1786814.

[26] S.S.h. Hosseini, A. Shahrjerdi, A. Vazifeshenas, Review of Relations for Physical Properties of Nanofluids, Australian Journal of Basic and Applied Sciences 5 (10) (2011) 417–435 5.

[27] M. Al-Maghalseh, K. Mahkamov, Methods of heat transfer intensification in PCM thermal storage systems: review paper, Renew. Sustain. Energy Rev. 92 (2018) 62–94, doi:10.1016/j.rser.2018.04.064.

[28] A. Sar, C. Alkan, D. Kahraman, A. Biçer, SMicro /nano-encapsulated n -heptadecane with polystyrene shell for latent heat thermal energy storage, Solar Energy Mater. Solar Cells 126 (2014) 42–50, doi:10.1016/j.solmat.2014.03.023.

[29] T. Gurmen Ozcelik, Preparation, characterization and thermal properties of paraffin wax–expanded perlite form-stable composites for latent heat storage, Mater. Sci. (Medziagotyra) 23 (1) (2017) 39–43, doi:10.5755/j01.ms.23.1.13661.

[30] F.S. Javadi, H.S.C. Metselaar, P. Ganesan, Performance improvement of solar thermal systems integrated with phase change materials (PCM), a review, Sol. Energy 206 (2020) 330–352, doi:10.1016/j.solener.2020.05.106.

[31] M. Mehrali, S.T. Latibari, M. Mehrali, H. Simon, C. Metselaar, Shape-stabilized phase change materials with high thermal conductivity based on paraffin /graphene oxide composite, Energy Convers. Manag. 67 (2013) 275–282, doi:10.1016/j.enconman.2012.11.023.

[32] P. Zhang, X. Xiao, Z.W. Ma, A review of the composite phase change materials: fabrication, characterization, mathematical modeling and application to performance enhancement, Appl. Energy 165 (2016) 472–510, doi:10.1016/j.apenergy.2015.12.043.

[33] K. Pielichowska, K. Pielichowski, Phase change materials for thermal energy storage, Prog. Mater. Sci. 65 (2014) 67–123, doi:10.1016/j.pmatsci.2014.03.005.

[34] C. Abeykoon, Single screw extrusion control: a comprehensive review and directions for improvements, Control Eng. Pract. 51 (2016) 69–80, doi:10.1016/j.conengprac.2016.03.008.

[35] C. Abeykoon, A.L. Kelly, E.C. Brown, P.D. Coates, The effect of materials, process settings and screw geometry on energy consumption and melt temperature in single screw extrusion, Appl. Energy 180 (2016) 880–894, doi:10.1016/j.apenergy.2016.07.014.

[36] H. Zhou, S. Zhang, M. Yang, The effect of heat-transfer passages on the effective thermal conductivity of high filler loading composite materials, Compos. Sci. Technol. 67 (6) (May 2007) 1035–1040, doi:10.1016/j.compscitech.2006.06.004.

[37] J. Wang, J.K. Carson, M.F. North, D.J. Cleland, A new approach to modelling the effective thermal conductivity of heterogeneous materials, Int. J. Heat Mass Transf. 49 (2006) 3075–3083, doi:10.1016/j.ijheatmasstransfer.2006.02.007.

[38] D. Stroud, The effective medium approximations: some recent developments, Superlatt. Microstruct. 23 (1998) 567–573, doi:10.1006/spmi.1997.0524.

[39] Y. Qu, S. Wang, D. Zhou, Y. Tian, Experimental study on thermal conductivity of paraffin-based shape-stabilized phase change material with hybrid carbon nano-additives, Renew. Energy 146 (2020) 2637–2645 February, doi:10.1016/j.renene.2019.08.098.

[40] R. Pal, On the Lewis – Nielsen model for thermal /electrical conductivity of composites, Compos. Part A: Appl. Sci. Manuf. 39 (2008) 718–726, doi:10.1016/j.compositesa.2008.02.008.

[41] J. Zeng, R. Fu, S. Agathopoulos, S. Zhang, X. Song, H. He, Numerical simulation of thermal conductivity of particle filled epoxy composites, J. Electron. Packag. 131 (2009) 1–7, doi:10.1115/1.4000210.

[42] X. Sun, Y. Chu, Y. Mo, S. Fan, S. Liao, P. Chu, et al., Experimental investigations on the heat transfer of melting phase change material (PCM), Energy Proc. 152 (2018) 186–191, doi:10.1016/j.egypro.2018.09.079.

[43] A. Sari, K. Kaygusuz, Thermal energy storage system using stearic acid as a phase change material, Sol. Energy 71 (2001) 365–376, doi:10.1016/S0038-092X(01)00075-5.

[44] A. Sari, K. Kaygusuz, Thermal performance of palmitic acid as a phase change energy storage material, Energy Convers. Manag. 43 (2002) 863–876, doi:10.1016/S0196-8904(01)00071-1.

[45] K.S. Hwang, J. Lee, S.P. Jang, Buoyancy-driven heat transfer of water-based Al2O3 nanofluids in a rectangular cavity, Int. J. Heat Mass Transf. 50 (2007) 4003–4010, doi:10.1016/j.ijheatmasstransfer.2007.01.037.

[46] R. Nasrin, M.A. Alim, A.J. Chamkha, International Buoyancy-driven heat transfer of water – Al 2 O 3 nanofluid in a closed chamber : effects of solid fraction, Prandtl number and aspect ratio, Int. J. Heat Mass Transf. 55 (2012) 7355–7365, doi:10.1016/j.ijheatmasstransfer.2012.08.011.

[47] A. Einstein, Investigations On the Theory of the Brownian Movement, Dover Publications, Inc., 1956.

[48] S. Jang, S. Choi, Role of Brownian motion in the enhanced thermal conductivity of nanofluids, Appl. Phys. Lett. 84 (2004) 4316, doi:10.1063/1.1756684.

[49] H.F. Oztop, E. Abu-nada, Numerical study of natural convection in partially heated rectangular enclosures filled with nanofluids, Int. J. Heat Fluid Flow 29 (2008) 1326–1336, doi:10.1016/j.ijheatfluidflow.2008.04.009.

[50] S.K. Das, S.U.S. Choi, H.E. Patel, Heat transfer in nanofluids–a review, Heat Transf. Eng. 27 (2006) 3–19, doi:10.1080/01457630600904593.

[51] X. Deng, Z. Huang, W. Wang, R.N. Davé, Investigation of nanoparticle agglomerates properties using Monte Carlo simulations, Adv. Powder Technol. 27 (2016) 1971–1979, doi:10.1016/j.apt.2016.06.029.

[52] D.R. Ceratti, M. Faustini, C. Sinturel, M. Vayer, V. Dahirel, M. Jardat, et al., Critical effect of pore characteristics on capillary infiltration in mesoporous films, Nanoscale 7 (2015) 5371–5382, doi:10.1039/c4nr03021d.

[53] Freedman J.P., Leach J.H., Preble E.A., Sitar Z., Davis R.F., Malen J.A. Universal phonon mean free path spectra in crystalline semiconductors at high temperature. Sci. Rep. 3, Article number: 2963 (2013). 10.1038/srep02963.

[54] D.G. Cahill, W.K. Ford, K.E. Goodson, G.D. Mahan, A. Majumdar, J. Humphrey, et al., Nanoscale thermal transport, J. Appl. Phys. 93 (2003) 793, doi:10.1063/1.1524305.

[55] J. Maire, R. Anufriev, R. Yanagisawa, A. Ramiere, S. Volz, M. Nomura, Heat conduction tuning by wave nature of phonons, Sci. Adv. 3 (8) (2017) 1–7 4 Aug, doi:10.1126/sciadv.1700027.

[56] D.L. Nika, A. Balandin, Two-dimensional phonon transport in graphene, J. Phys. Condens. Matter 24 (23) (2012), doi:10.1088/0953-8984/24/23/233203.

[57] S. Ghosh, W. Bao, D.L. Nika, S. Subrina, E.P. Pokatilov, C.N. Lau, et al., Dimensional crossover of thermal transport in few-layer graphene, Nat. Mater. 9 (2010) 555–558, doi:10.1038/nmat2753.

[58] B. Wang, J. Wu, X. Gu, H. Yin, Y. Wei, R. Yang, M. Dresselhaus, Stable planar single-layer hexagonal silicene under tensile strain and its anomalous Poisson's ratio, Appl. Phys. Lett. 104 (2014) 081902, doi:10.1063/1.4866415.

[59] X. Li, K. Maute, M.L. Dunn, R. Yang, Strain effects on the thermal conductivity of nanostructures, Phys. Rev. B 81 (245318) (2010) 1–11, doi:10.1103/PhysRevB.81.245318.

[60] J.M. Ziman, Electrons and Phonons, Oxford University Press, London, 1960.

[61] Lindsay L., Broido D.A., Mingo N. Flexural phonons and thermal transport in multilayer graphene and graphite 2011;235428:1–5. 10.1103/PhysRevB.83.235428.

[62] Lindsay L., Broido D.A., Mingo N. Flexural phonons and thermal transport in graphene 2010:2–7. 10.1103/PhysRevB.82.115427.

[63] S. Hida, T. Hori, T. Shiga, J. Elliott, J. Shiomi, Thermal resistance and phonon scattering at the interface between carbon nanotube and amorphous polyethylene, Int. J. Heat Mass Transf. 67 (2013) 1024–1029, doi:10.1016/j.ijheatmasstransfer.2013.08.068.

[64] Shtein M., Nadiv R., Buzaglo M., Kahil K., Regev O. Thermally conductive graphene-polymer composites: size, percolation, and synergy effects 2015. 10.1021/cm504550e.

[65] L. Zeng, K.C. Collins, Y. Hu, M.N. Luckyanova, A.A. Maznev, S. Huberman, et al., Measuring phonon mean free path distributions by probing quasiballistic phonon transport in grating nanostructures, Sci. Rep. 5 (2015) 1–10, doi:10.1038/srep17131.

[66] Guo Z., Zhang D., Gong X. Thermal conductivity of graphene nanoribbons 2011;163103. 10.1063/1.3246155.

[67] N. Delibozov, Analysis of graphene nanoribbons passivated with gold, copper and indium, Int. J. Theor. Appl. Nanotechnol. 1 (2013) 5–10, doi:10.11159/ijtan.2013.005.

[68] Phys J.A. Thermal conductivity of GaN single crystals : influence of impurities incorporated in different growth processes 2018;105106. 10.1063/1.5047531.

[69] T.M. Gibbons, M.B. Bebek, B.y. Kang, C.M. Stanley, S.K. Estreicher, Phonon-phonon interactions : first principles theory, J. Appl. Phys. 118 (2015) 085103, doi:10.1063/1.4929452.

[70] T. Evgin, H. Dogacan, N. Horny, A. Turgut, I. Hakkı, Effect of aspect ratio on thermal conductivity of high density polyethylene /multi-walled carbon nanotubes nanocomposites, Compos. Part A: Appl. Sci. Manuf. 82 (2016) 208–213, doi:10.1016/j.compositesa.2015.12.013.

[71] K.M.F. Shahil, A.A. Balandin, Graphene–multilayer graphene nanocomposites as highly efficient thermal interface materials, Nano Lett. 12 (2) (2012) 861–867, doi:10.1021/nl203906r.

[72] A. Yu, P. Ramesh, M.E. Itkis, E. Bekyarova, R.C. Haddon, Graphite nanoplatelet–epoxy composite thermal interface materials, J. Phys. Chem. C 111 (21) (2007) 7565–7569, doi:10.1021/jp071761s.

[73] S. Wu, R.B. Ladani, J. Zhang, E. Bafekrpour, K. Ghorbani, A.P. Mouritz, et al., Aligning multilayer graphene flakes with an external electric field to improve multifunctional properties of epoxy nanocomposites, Carbon N Y 94 (2015) 607–618, doi:10.1016/j.carbon.2015.07.026.

[74] H. Babaei, P. Keblinski, J.M. Khodadadi, International Thermal conductivity enhancement of paraffins by increasing the alignment of molecules through adding CNT /graphene q, Int. J. Heat Mass Transf. 58 (2013) 209–216, doi:10.1016/j.ijheatmasstransfer.2012.11.013.

[75] F. Kargar, Z. Barani, J.S. Lewis, B. Debnath, R. Salgado, R. Lake, et al., Thermal percolation threshold and thermal properties of composites with graphene and boron nitride fillers, ACS Appl. Mater. Interfaces 10 (43) (2018 Oct 31) 37555–37565, doi:10.1021/acsami.8b16616.

[76] N. Putra, M. Amin, E.A. Kosasih, R.A. Luanto, N.A. Abdullah, Characterization of the thermal stability of RT 22 HC /graphene using a thermal cycle method based on thermoelectric methods, Appl. Therm. Eng. 124 (2017) 62–70, doi:10.1016/j.applthermaleng.2017.06.009.

[77] Y. Liu, D. Zhang, Effect of covalent functionalization and phase change matrix on heat transfer across graphene /phase change material interfaces, Appl. Therm. Eng. 151 (2019) 38–45, doi:10.1016/j.applthermaleng.2019.02.002.

[78] J.F. Li, W. Lu, Y.B. Zeng, Z.P. Luo, Simultaneous enhancement of latent heat and thermal conductivity of docosane-based phase change material in the presence of spongy graphene, Sol. Energy Mater Sol. Cells 128 (2014) 48–51, doi:10.1016/j.solmat.2014.05.018.

[79] Zhao Y., Hu C., Hu Y., Cheng H., Shi G., Q.u. L. A. Versatile, Ultralight, Nitrogen-Doped Graphene Framework Angewandte Chemie International Edition Volume 51, Issue 45 p. 11371-11375 doi:10.1002/anie.201206554.

[80] W. Li, Y. Dong, J. Wang, Y. Zhang, X. Zhang, X. Liu, Experimental study on enhanced heat transfer of nanocomposite phase change materials, Phase Transit. 92 (3) (2019), doi:10.1080/01411594.2019.1574010.

[81] Mishra A.K., Lahiri B.B., Philip J. Effect of surface functionalization and physical properties of nanoinclusions on thermal conductivity enhancement in an organic phase change material 2018. 10.1021/acsomega.8b01084.

[82] L.A. Richards, Capillary conduction of liquids through porous mediums, J. Appl. Phys. 1 (1931) 318–333, doi:10.1063/1.1745010.

[83] T. Nomura, N. Okinaka, T. Akiyama, Impregnation of porous material with phase change material for thermal energy storage, Mater. Chem. Phys. 115 (2009) 846–850, doi:10.1016/j.matchemphys.2009.02.045.

[84] S. Ranjbar, H. Masoumi, R. Haghighi, K. Mojtaba, Experimental investigation of stability and thermal conductivity of phase change materials containing pristine and functionalized multi – walled carbon nanotubes, J. Therm. Anal. Calorim. 140 (2020) 2505–2518, doi:10.1007/s10973-019-09005-x.

[85] Bauer T., Steinmann W., Laing D., Tamme R. Thermal energy storage materials and systems. Ann. Rev. Heat Transf., 15, p.131–177 10.1615/AnnualRevHeat-Transfer.2012004651

[86] Functional textiles for improved performance, protection and health 2011. 10.1533/9780857092878

[87] M.M. Farid, A.M. Khudhair, S. Ali, K. Razack, A review on phase change energy storage : materials and applications, Energy Convers. Manag. 45 (June 2004) 1597–1615 Issues 9–10, doi:10.1016/j.enconman.2003.09.015.

[88] M. Veerappan, S. Kalaiselvam, S. Iniyan, R. Goic, Phase change characteristic study of spherical PCMs in solar energy storage, Sol. Energy 83 (2009) 1245–1252, doi:10.1016/j.solener.2009.02.006.

[89] Hasnain S.M. Review on sustainable thermal energy storage technologies, Part I: heat storage materials and techniques Energy Convers. Manag., 39, (11), 1 August 1998, 1127–1138. 10.1016/S0196-8904(98)00025-9

[90] N. Sarier, E. Onder, The manufacture of microencapsulated phase change materials suitable for the design of thermally enhanced fabrics, Thermochim. Acta 452 (2) (2007) 149–160 15 January, doi:10.1016/j.tca.2006.08.002.

[91] A. Abhat, Low temperature latent heat thermal energy storage:, Heat Storage Mater. Solar Energy 30 (4) (1983) 313–332, doi:10.1016/0038-092X(83)90186-X.

[92] D. Feldman, M.M. Shapiro, D. Banu, Organic phase change materials for thermal energy storage, Sol. Energy Mater. 13 (1986) 1–10.

[93] Feldman, D., Shapiro, M.M., and Banu D.C.J. Fuks solar energy materials 18 (1989) 201–216.

[94] Ş. Nurten, M. Fois, H. Paksoy, Improving thermal conductivity phase change materials–a study of paraffin nanomagnetite composites, Solar Energy Mater. Sol. Cells 137 (2015) 61–63.

[95] Y. Lin, G. Alva, G. Fang, Review on thermal performances and applications of thermal energy storage systems with inorganic phase change materials, Energy 165 (2018) 685–708, doi:10.1016/j.energy.2018.09.128.

[96] S.A. Mohamed, F.A. Al-sulaiman, N.I. Ibrahim, H. Zahir, A. Al-ahmed, R. Saidur, et al., A review on current status and challenges of inorganic phase change materials for thermal energy storage systems, Renew. Sustain. Energy Rev. 70 (2017) 1072–1089 April, doi:10.1016/j.rser.2016.12.012.

[97] A. Shanker, C. Li, G. Kim, D. Gidley, K.P. Pipe, J. Kim, High thermal conductivity in electrostatically engineered amorphous polymers, Sci. Adv. 3 (7)) (2017) 28 Jul, doi:10.1126/sciadv.1700342.

[98] C. Abeykoon, P. Sri-Amphorn, A. Fernando, Optimization of fused deposition modeling parameters for improved PLA and ABS 3D printed structures, Int. J. Light Mater. Manuf. 3 (2020) 284–297, doi:10.1016/j.ijlmm.2020.03.003.

[99] Y. Yuan, C. Abeykoon, W. Mirihanage, A. Fernando, Y.C. Kao, J.A.W. Harings, Prediction of temperature and crystal growth evolution during 3D printing of polymeric materials via extrusion, Mater. Des. 196 (2020) 109121, doi:10.1016/j.matdes.2020.109121.

[100] R. Rao, C.L. Pint, A.E. Islam, R.S. Weatherup, S. Hofmann, E.R. Meshot, et al., Carbon nanotubes and related nanomaterials: critical advances and challenges for synthesis toward mainstream commercial applications, ACS Nano 12 (2018) 11756–11784, doi:10.1021/acsnano.8b06511.

[101] G. Kucinskis, G. Bajars, J. Kleperis, Graphene in lithium ion battery cathode materials: a review, J. Power Sources 240 (2013) 66–79, doi:10.1016/j.jpowsour.2013.03.160.

[102] A.E. Agboola, R.W. Pike, T.A. Hertwig, H.H. Lou, Conceptual design of carbon nanotube processes, Clean Technol. Environ. Policy 9 (2007) 289–311, doi:10.1007/s10098-006-0083-2.

[103] https://www.cheaptubes.com/product-category/multi-walled-carbon-nanotubes/. Multi walled carbon nanotubes products (Accessed September 2021).

[104] G. Xin, H. Sun, S.M. Scott, T. Yao, F. Lu, D. Shao, et al., Advanced phase change composite by thermally annealed defect-free graphene for thermal energy storage, ACS Appl. Mater. Interfaces 6 (2014) 15262–15271, doi:10.1021/am503619a.

[105] Q. Zhang, J. Liu, Anisotropic thermal conductivity and photodriven phase change composite based on RT100 in fi ltrated carbon nanotube array, Sol. Energy Mater. Sol. Cells 190 (2019) 1–5, doi:10.1016/j.solmat.2018.10.010.

[106] Z. Ling, J. Chen, T. Xu, X. Fang, X. Gao, Z. Zhang, Thermal conductivity of an organic phase change material /expanded graphite composite across the phase change temperature range and a novel thermal conductivity model q, Energy Convers. Manag. 102 (2015) 202–208, doi:10.1016/j.enconman.2014.11.040.

[107] L. Xia, P. Zhang, R.Z. Wang, Preparation and thermal characterization of expanded graphite /paraffin composite phase change material, Carbon N Y 48 (2010) 2538–2548, doi:10.1016/j.carbon.2010.03.030.

[108] C.X. Guo, X. Ma, L. Yang, PCM /graphite foam composite for thermal energy storage device PCM / graphite foam composite for thermal energy storage device, IOP Conf. Ser.: Mater. Sci. Eng. 87 (01) (2014), doi:10.1088/1757-899X/87/1/012014.

[109] L. Chen, R. Zou, W. Xia, Z. Liu, Y. Shang, J. Zhu, et al., Electro- and Photodriven phase change composites based on wax-infiltrated carbon nanotube sponges, ACS Nano (2012) 10884–10892, doi:10.1021/nn304310n.

[110] M. Karthik, A. Faik, B.D. Aguanno, Preparation of erythritol – graphite foam phase change composite with enhanced thermal conductivity for thermal energy storage applications, Carbon N Y 94 (2015) 266–276, doi:10.1016/j.carbon.2015.06.075.

[111] Q. Cao, F. He, C. Xie, J. Fan, J. Wu, Paraffin-based shape-stable phase change materials with graphene /carbon nanotube three- dimensional network structure. Fullerenes, Nanotub. Carbon Nanostruct. 27 (2019) 492–497, doi:10.1080/1536383X.2019.1611562.

[112] X. Xiao, P. Zhang, Morphologies and thermal characterization of paraf fi n /carbon foam composite phase change material, Sol. Energy Mater. Sol. Cells 117 (2013) 451–461, doi:10.1016/j.solmat.2013.06.037.

[113] D.G. Atinafu, C. Wang, W. Dong, X. Chen, M. Du, H. Gao, et al., Cells *in-situ* derived graphene from solid sodium acetate for enhanced photothermal conversion, thermal conductivity, and energy storage capacity of phase change materials, Sol. Energy Mater. Sol. Cells 205 (2020) 110269, doi:10.1016/j.solmat.2019.110269.

[114] X. Fang, P. Hao, B. Song, C. Tuan, C. Wong, Z. Yu, Form-stable phase change material embedded with chitosan-derived carbon aerogel, Mater. Lett. 195 (2017) 79–81, doi:10.1016/j.matlet.2017.02.075.

[115] Qiu J., Fan X., Shi Y., Zhang S. PEG/3D graphene oxide network form-stable phase change materials with ultrahigh filler content 2019:21371–7. 10.1039/c9ta07629h.

[116] R. Singh, S. Sadeghi, B. Shabani, Thermal conductivity enhancement of phase change materials for low-temperature thermal energy storage applications, Energies 12 (2019), doi:10.3390/en12010075.

[117] P.A. Hartley, G.D. Parfitt, L.B. Pollack, The role of the van der Waals force in the agglomeration of powders containing submicron particles, Powder Technol. 42 (1) (1985) 35–46 April, doi:10.1016/0032-5910(85)80036-X.

[118] L. Sun, B. Fugetsu, Mass production of graphene oxide from expanded graphite, Mater Lett. 109 (2013) 207–210, doi:10.1016/j.matlet.2013.07.072.

[119] M. Almeida, Waste materials for activated carbon preparation and its use in aqueous- phase treatment : a review, J. Environ. Manage. 85 (4) (2007) 833–846 December, doi:10.1016/j.jenvman.2007.07.031.

[120] Z. Chen, F. Shan, L. Cao, G. Fang, Synthesis and thermal properties of shape-stabilized lauric acid /activated carbon composites as phase change materials for thermal energy storage, Sol. Energy Mater. Sol. Cells 102 (2012) 131–136, doi:10.1016/j.solmat.2012.03.013.

[121] S.I. Hussain, R. Dinesh, A.A. Roseline, S. Dhivya, S. Kalaiselvam, Enhanced thermal performance and study the influence of sub cooling on activated carbon dispersed eutectic PCM for cold storage applications, Energy Build. 143 (15 May 2017) 17–24.

[122] J. Yang, X. Li, S. Han, Y. Zhang, P. Min, N. Koratkar, et al., Air-dried, high-density graphene hybrid aerogels for phase change composites with exceptional thermal conductivity and shape stability, J. Mater. Chem. A 4 (2016) 18067–18074, doi:10.1039/c6ta07869a.

[123] J. Yang, E. Zhang, X. Li, Y. Zhang, J. Qu, Z. Yu, Cellulose /graphene aerogel supported phase change composites with high thermal conductivity and good shape stability for thermal energy storage, Carbon N Y 98 (2016) 50–57, doi:10.1016/j.carbon.2015.10.082.

[124] L. Shao, L. Shi, X. Li, N. Song, P. Ding, Synergistic effect of BN and graphene nanosheets in 3D framework on the enhancement of thermal conductive properties of polymeric composites, Compos. Sci. Technol. 135 (2016) 83–91, doi:10.1016/j.compscitech.2016.09.013.

[125] M. Wang, T. Zhang, D. Mao, Y. Yao, X. Zeng, L. Ren, et al., Highly compressive boron nitride nanotube aerogels reinforced with reduced graphene oxide, ACS Nano (2019), doi:10.1021/acsnano.9b03225.

[126] Y. Cao, D. Fan, S. Lin, L. Mu, F.T.T. Ng, Q. Pan, Phase change materials based on comb-like polynorbornenes and octadecylamine-functionalized graphene oxide nanosheets for thermal energy storage, Chem. Eng. J. 389 (2020) 124318, doi:10.1016/j.cej.2020.124318.

[127] X. Li, Y. Chen, S. Mo, L. Jia, X. Shao, Effect of surface modi fi cation on the stability and thermal conductivity of water-based SiO_2 -coated graphene nano fluid, Thermochim. Acta 595 (2014) 6–10, doi:10.1016/j.tca.2014.09.006.

[128] S. Li, L. Kong, H. Wang, H. Xu, J. Li, H. Shi, Thermal performance and shape-stabilization of comb-like polymeric phase change materials enhanced by octadecylamine-functionalized graphene oxide, Energy Convers. Manag. 168 (2018) 119–127, doi:10.1016/j.enconman.2018.05.014.

[129] P. Goli, S. Legedza, A. Dhar, R. Salgado, J. Renteria, A.A. Balandin, Graphene-enhanced hybrid phase change materials for thermal management of Li-ion batteries, J. Power Sources 248 (2014) 37–43, doi:10.1016/j.jpowsour.2013.08.135.

[130] S. Wu, T.X. Li, T. Yan, Y.J. Dai, R.Z. Wang, High performance form-stable expanded graphite /stearic acid composite phase change material for modular thermal energy storage, Int. J. Heat Mass Transf. 102 (2016) 733–744, doi:10.1016/j.ijheatmasstransfer.2016.06.066.

[131] R.J. Warzoha, A.S. Fleischer, Improved heat recovery from paraffin-based phase change materials due to the presence of percolating graphene networks, Int. J. Heat Mass Transf. 79 (2014) 314–323, doi:10.1016/j.ijheatmasstransfer.2014.08.009.

[132] G. Ma, J. Sun, Y. Zhang, Y. Jing, Y. Jia, Preparation and thermal properties of stearic acid-benzamide eutectic mixture /expanded graphite composites as phase change materials for thermal energy storage, Powder Technol. 342 (2019) 131–140, doi:10.1016/j.powtec.2018.09.074.

[133] M. Amin, N. Putra, E.A. Kosasih, E. Prawiro, R. Achmad, T.M.I. Mahlia, Thermal properties of beeswax /graphene phase change material as energy storage for building applications, Appl. Therm. Eng. 112 (2017) 273–280, doi:10.1016/j.applthermaleng.2016.10.085.

[134] A.K. Mishra, B.B. Lahiri, V. Solomon, J. Philip, Nano-inclusion aided thermal conductivity enhancement in palmitic acid/di-methyl formamide phase

change material for latent heat thermal energy storage, Thermochim. Acta 678 (2019) 178309, doi:10.1016/j.tca.2019.178309.
[135] A. Zabalegui, D. Lokapur, H. Lee, Nanofluid PCMs for thermal energy storage : latent heat reduction mechanisms and a numerical study of effective thermal storage performance, Int. J. Heat Mass Transf. 78 (2014) 1145–1154, doi:10.1016/j.ijheatmasstransfer.2014.07.051.
[136] Y. Li, Y. Li, X. Huang, H. Zheng, G. Lu, Z. Xi, et al., Graphene-CoO/PEG composite phase change materials with enhanced solar-to-thermal energy conversion and storage capacity, Compos. Sci. Technol. 195 (2020) 108197, doi:10.1016/j.compscitech.2020.108197.
[137] F. Wang, P. Zhang, Y. Mou, M. Kang, M. Liu, L. Song, et al., Synthesis of the polyethylene glycol solid-solid phase change materials with a functionalized graphene oxide for thermal energy storage, Polym. Test. 63 (2017) 494–504 October, doi:10.1016/j.polymertesting.2017.09.005.
[138] W. Du, Z. Zhang, H. Su, H. Lin, Z. Li, Urethane-Functionalized graphene oxide for improving compatibility and thermal conductivity of waterborne polyurethane composites, Ind. Eng. Chem. Res. 57 (2018) 7146–7155, doi:10.1021/acs.iecr.8b00656.
[139] A.R. Akhiani, M. Mehrali, S.T. Latibari, M. Mehrali, T. Meurah, I. Mahlia, et al., One-step preparation of form-stable phase change material through self-assembly of fatty acid and graphene, J. Phys. Chem. C (2015), doi:10.1021/acs.jpcc.5b06089.
[140] J. Ge, Y. Wang, H. Wang, H. Mao, J. Li, H. Shi, Thermal properties and shape stabilization of epoxidized methoxy polyethylene glycol composite PCMs tailored by polydopamine-functionalized graphene oxide, Sol. Energy Mater. Sol. Cells 208 (2020) 110388, doi:10.1016/j.solmat.2019.110388.
[141] W. Wnag, B. Tang, Fe3O4-functionalized graphene nanosheet embedded phase change material composites: efficient magnetic- and sunlight-driven energy conversion and storage, J. Mater. Chem. A 5 (2017) 958–968, doi:10.1039/c6ta07144a.
[142] Z. Li, R.J. Young, R. Wang, F. Yang, L. Hao, W. Jiao, et al., The role of functional groups on graphene oxide in epoxy nanocomposites, Polymer (Guildf) 54 (2013) 5821–5829, doi:10.1016/j.polymer.2013.08.026.
[143] Y. Ahmadi, K.-.H. Kim, S. Kim, M. Tabatabaei, Recent advances in polyurethanes as efficient media for thermal energy storage, Energy Storage Mater. 30 (2020) 74–86, doi:10.1016/j.ensm.2020.05.003.
[144] M. Mohammad, I. Jo, L. Shi, Phonon-interface scattering in multilayer graphene on an amorphous support, PNAS 8 (41) (2013) 16321–16326 October110, doi:10.1073/pnas.1306175110.
[145] Y. Zhou, C. Li, H. Wu, S. Guo, Construction of hybrid graphene oxide /graphene nanoplates shell in para ffi n microencapsulated phase change materials to improve thermal conductivity for thermal energy storage, Colloids Surf. A 597 (2020) 124780, doi:10.1016/j.colsurfa.2020.124780.
[146] T.D. Dao, H. Mo, Novel stearic acid /graphene core – shell composite microcapsule as a phase change material exhibiting high shape stability and performance, Sol. Energy Mater. Sol. Cells 137 (2015) 227–234, doi:10.1016/j.solmat.2015.02.009.
[147] V.V. Tyagi, S.C. Kaushik, S.K. Tyagi, T. Akiyama, Development of phase change materials based microencapsulated technology for buildings : a review, Renew. Sustain. Energy Rev. 15 (2011) 1373–1391, doi:10.1016/j.rser.2010.10.006.
[148] E.M. Shchukina, M. Graham, Z. Zheng, D.G. Shchukin, Nanoencapsulation of phase change materials for advanced thermal energy storage systems, Chem. Soc. Rev. (2018) 4156–4175, doi:10.1039/c8cs00099a.
[149] D. Cai, K. Yusoh, M. Song, The mechanical properties and morphology of a graphite oxide nanoplatelet /polyurethane composite, Nanotechnology 20 (2009) 085712, doi:10.1088/0957-4484/20/8/085712.
[150] Y. Zhou, C. Li, H. Wu, S. Guo, Construction of hybrid graphene oxide /graphene nanoplates shell in para ffi n microencapsulated phase change materials to improve thermal conductivity for thermal energy storage, Colloids Surf. A 597 (2020) 124780, doi:10.1016/j.colsurfa.2020.124780.
[151] R.D. Weinstein, T.C. Kopec, A.S. Fleischer, E. D'Addio, C.A. Bessel, The experimental exploration of embedding phase change materials with graphite nanofibers for the thermal management of electronics, J. Heat Transf. 130 (2008) 1–8, doi:10.1115/1.2818764.
[152] S. Harish, K. Ishikawa, S. Chiashi, J. Shiomi, S. Maruyama, Anomalous thermal conduction characteristics of phase change composites with single-walled carbon nanotube inclusions, J. Phys. Chem. C 117 (2013) 15409–15413, doi:10.1021/jp4046512.
[153] X. Zhu, L. Han, Y. Lu, F. Wei, X. Jia, Geometry-induced thermal storage enhancement of shape-stabilized phase change materials based on oriented carbon nanotubes, Appl. Energy 254 (2019) 113688, doi:10.1016/j.apenergy.2019.113688.
[154] X. Zhu, L. Han, Y. Lu, F. Wei, X. Jia, Geometry-induced thermal storage enhancement of shape-stabilized phase change materials based on oriented carbon nanotubes, Appl. Energy 254 (2019) 113688, doi:10.1016/j.apenergy.2019.113688.
[155] Yang L., Zhang N., Yuan Y. Thermal performance of stearic acid /carbon nanotube composite phase change materials for energy storage prepared by ball milling 2019:8–10. 10.1002/er.4352.
[156] Y. Cui, C. Liu, S. Hu, X. Yu, The experimental exploration of carbon nanofiber and carbon nanotube additives on thermal behavior of phase change materials, Sol. Energy Mater Sol. Cells 95 (2011) 1208–1212, doi:10.1016/j.solmat.2011.01.021.
[157] S. Salyan, S. Suresh, Multi-walled carbon nanotube laden with D-Mannitol as phase change material: characterization and experimental investigation, Adv. Powder Technol. 29 (2018) 3183–3191, doi:10.1016/j.apt.2018.08.021.
[158] W. Xuehong, W. Chunxu, W. Yanling, Z. Youjian, Experimental study of thermo-physical properties and application of paraffin-carbon nanotubes composite phase change materials, Int. J. Heat Mass Transf. 140 (2019) 671–677, doi:10.1016/j.ijheatmasstransfer.2019.06.008.
[159] C. Yadav, R. Rekha Sahoo, Experimental analysis for optimum thermal performance and thermophysical parameters of MWCNT based capric acid PCM by using T-history method, Powder Technol. 364 (2020) 392–403, doi:10.1016/j.powtec.2020.02.008.
[160] R. Agrawal, J. Hanna, I.E. Gunduz, C.C. Luhrs, Epoxy–PCM composites with nanocarbons or multidimensional boron nitride as heat flow enhancers, Molecules (10) (2019) 1883 May; 24, doi:10.3390/molecules24101883.
[161] R. Cao, S. Chen, Y. Wang, N. Han, H. Liu, Functionalized carbon nanotubes as phase change materials with enhanced thermal, electrical conductivity, light-to-thermal, and electro-to-thermal performances, Carbon N Y 149 (2019) 263–272, doi:10.1016/j.carbon.2019.04.005.
[162] Science E. Characterization of shape-stabilized phase change material using beeswax and functionalized multi-walled carbon nanotubes. IOP Conf. Ser.: Earth Environ. Sci. 2018 105 012042. 10.1088/1755-1315/105/1/012042
[163] C. Wang, K. Chen, J. Huang, Z. Cai, Z. Hu, T. Wang, Thermal behavior of polyethylene glycol based phase change materials for thermal energy storage with multiwall carbon nanotubes additives, Energy 180 (2019) 873–880, doi:10.1016/j.energy.2019.05.163.
[164] X. Meng, S. Qin, H. Fan, Z. Huang, J. Hong, X. Xu, et al., Long alkyl chain-grafted carbon nanotube-decorated binary-core phase-change microcapsules for heat energy storage : synthesis and thermal properties, Sol. Energy Mater Sol. Cells 212 (2020) 110589, doi:10.1016/j.solmat.2020.110589.
[165] R. Zheng, J. Gao, J. Wang, G. Chen, Reversible temperature regulation of electrical and thermal conductivity using liquid–solid phase transitions, Nat. Commun. (2011), doi:10.1038/ncomms1288.
[166] Yu B.A., Ramesh P., Sun X., Bekyarova E., Itkis M.E., Haddon R.C.. Enhanced thermal conductivity in a hybrid graphite nanoplatelet–carbon nanotube filler for epoxy composites. Adv. Mater., 20, (24 p. 4740–4744 10.1002/adma.200800401.
[167] Y. Chen, Y. Liu, Z. Chen, Q. Shi, J. Gao, H. Ding, et al., Efficient shape-stabilized phase-change material based on novel mesoporous carbon microspheres as a matrix for polyethylene glycol: preparation and thermal properties, Jom 71 (2019) 4547–4555, doi:10.1007/s11837-019-03787-z.